Physical Chemistry (I)

Lecture 19

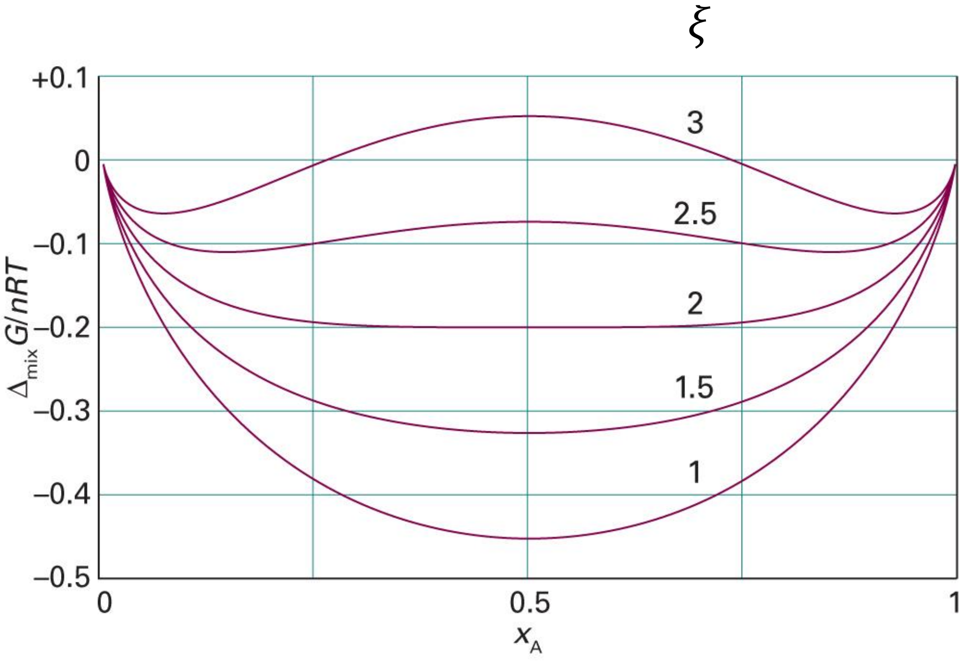
Special Lecture on

Biomolecular Phase Separation

Review

BW Model: Gibbs Energy of Mixing

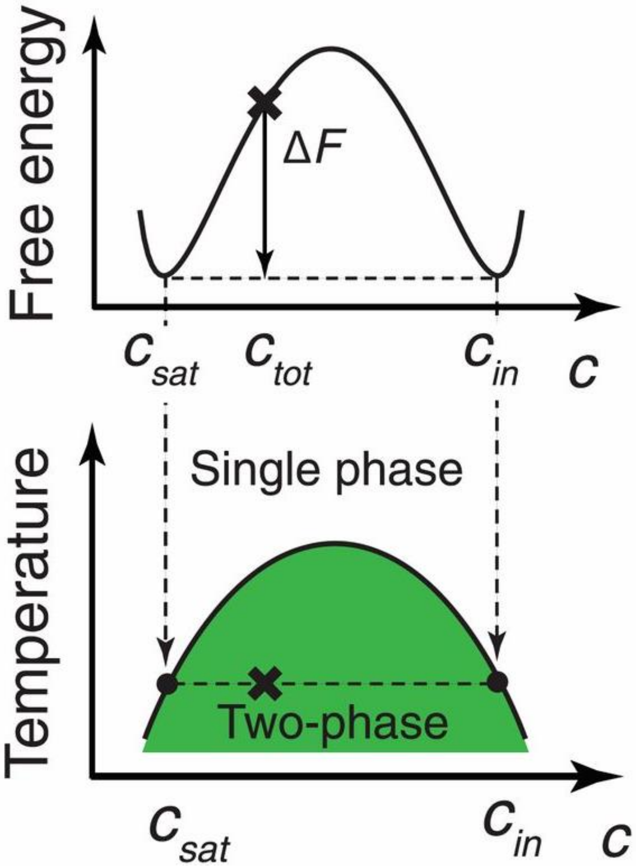
$$\Delta_{\text{mix}}G = nRT(x_A \ln x_A + x_B \ln x_B + \xi x_A x_B)$$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5B.5)

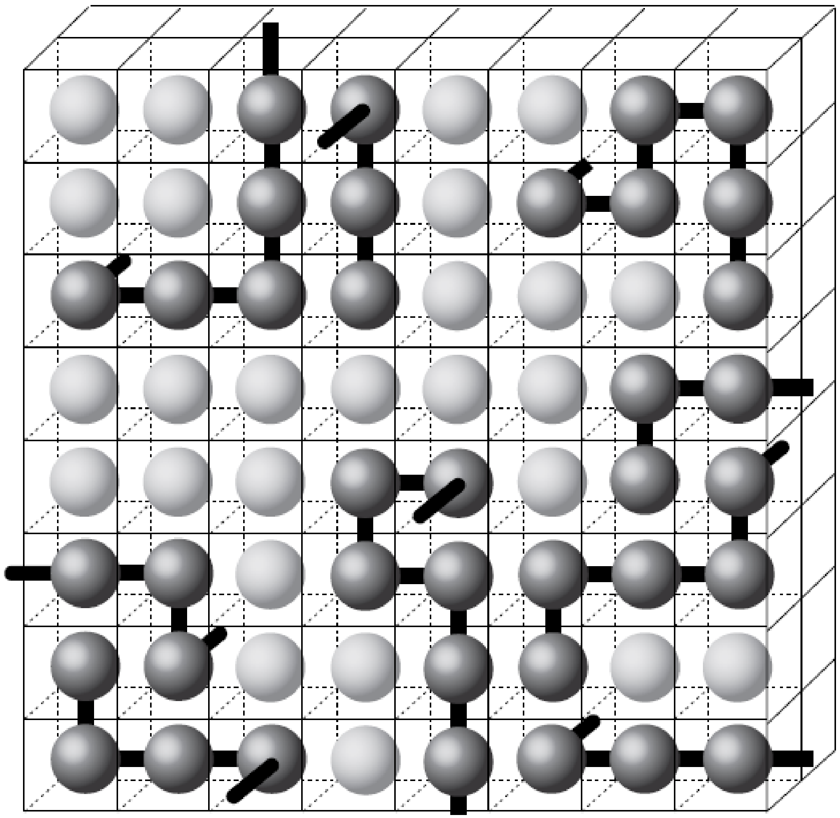
Review

What Happens at $\xi > 2$?



(Image: Shin and Brangwynne, *Science* 357: eaaf4382 (2017), Figure 3)

Lattice Model of Polymers



M sites

N_s solvent molecules

N_p polymer molecules

L monomers / polymer

(Image: Dill and Bromberg, *Molecular Driving Forces*, 2nd Ed., Figures 32.4)

Entropy of Mixing

$$\Delta_{\mathrm{mix}} S = -k_{\mathrm{B}} \left(N_{\mathrm{s}} \ln \frac{N_{\mathrm{s}}}{M} + N_{\mathrm{p}} \ln \frac{L N_{\mathrm{p}}}{M} \right)$$

Define volume fractions:

$$\phi_{\mathrm{s}} = \frac{N_{\mathrm{s}}}{M}, \qquad \phi_{\mathrm{p}} = \frac{L N_{\mathrm{p}}}{M}$$

Hence,

$$\frac{\Delta_{\mathrm{mix}} S}{M} = -k_{\mathrm{B}} \left(\phi_{\mathrm{s}} \ln \phi_{\mathrm{s}} + \frac{\phi_{\mathrm{p}}}{L} \ln \phi_{\mathrm{p}} \right)$$

Thermodynamics of Mixing

- Entropy of mixing

$$\frac{\Delta_{\mathrm{mix}} S}{M} = -k_{\mathrm{B}} \left(\phi_{\mathrm{s}} \ln \phi_{\mathrm{s}} + \frac{\phi_{\mathrm{p}}}{L} \ln \phi_{\mathrm{p}} \right)$$

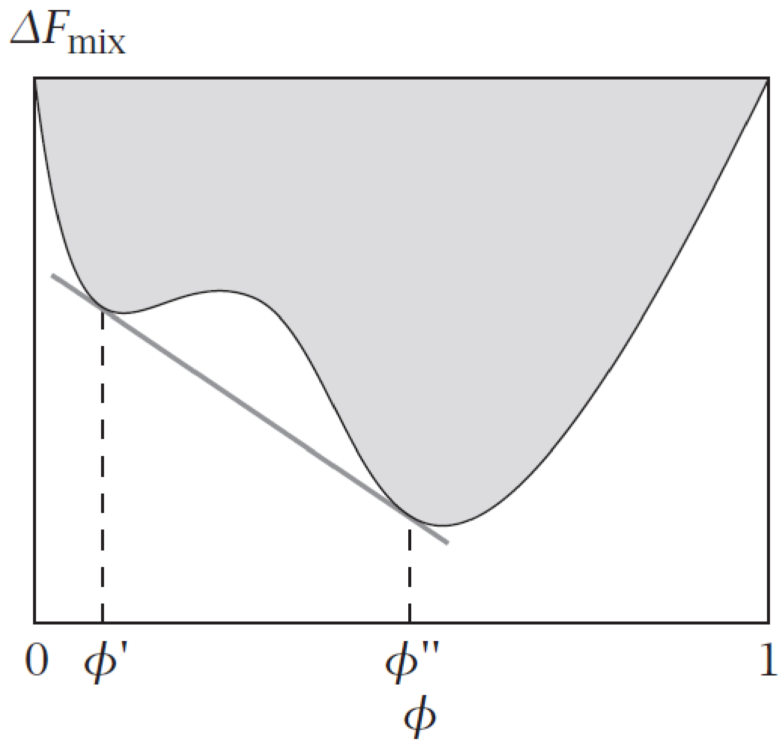
- Energy of mixing

$$\Delta_{\mathrm{mix}} U = k_{\mathrm{B}} T \xi_{\mathrm{sp}} \frac{N_{\mathrm{s}} N_{\mathrm{p}} L}{M} = M k_{\mathrm{B}} T \xi_{\mathrm{sp}} \phi_{\mathrm{s}} \phi_{\mathrm{p}}$$

- Helmholtz energy of mixing

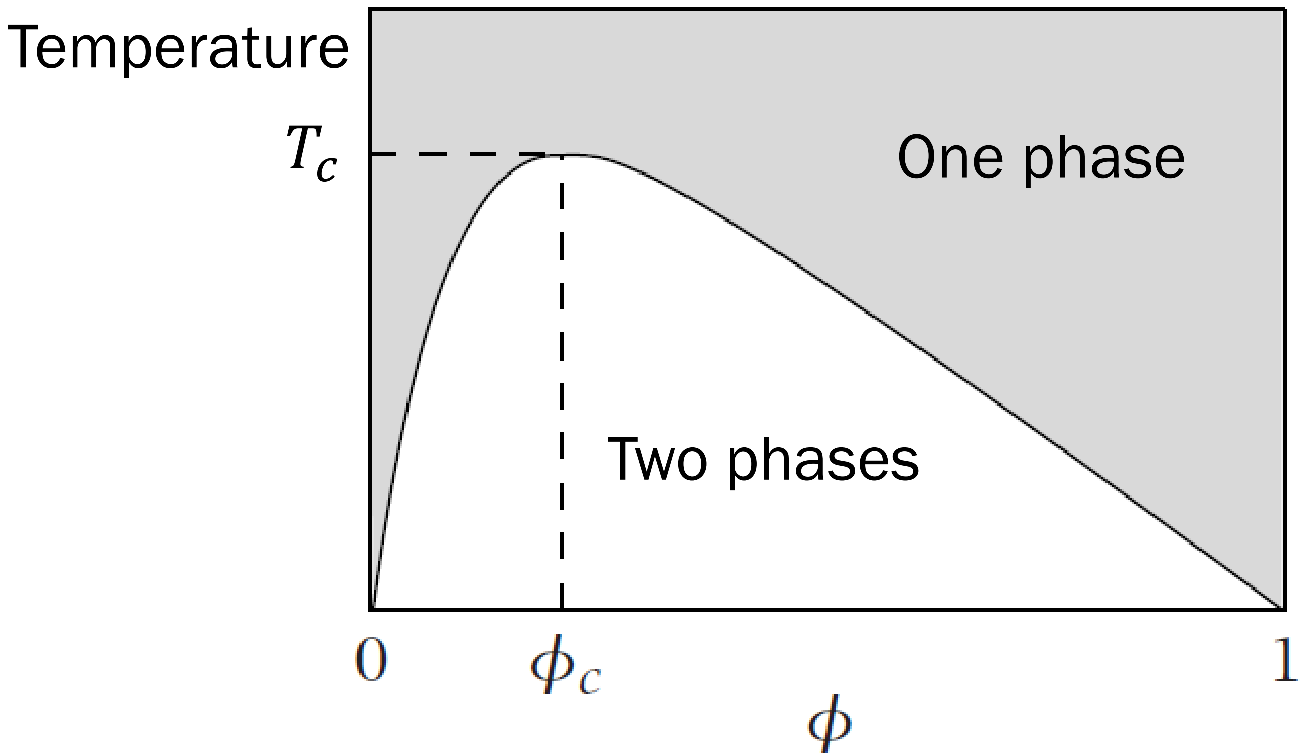
$$\frac{\Delta_{\mathrm{mix}} F}{M k_{\mathrm{B}} T} = \phi_{\mathrm{s}} \ln \phi_{\mathrm{s}} + \frac{\phi_{\mathrm{p}}}{L} \ln \phi_{\mathrm{p}} + \xi_{\mathrm{sp}} \phi_{\mathrm{s}} \phi_{\mathrm{p}}$$

Free Energy of Mixing



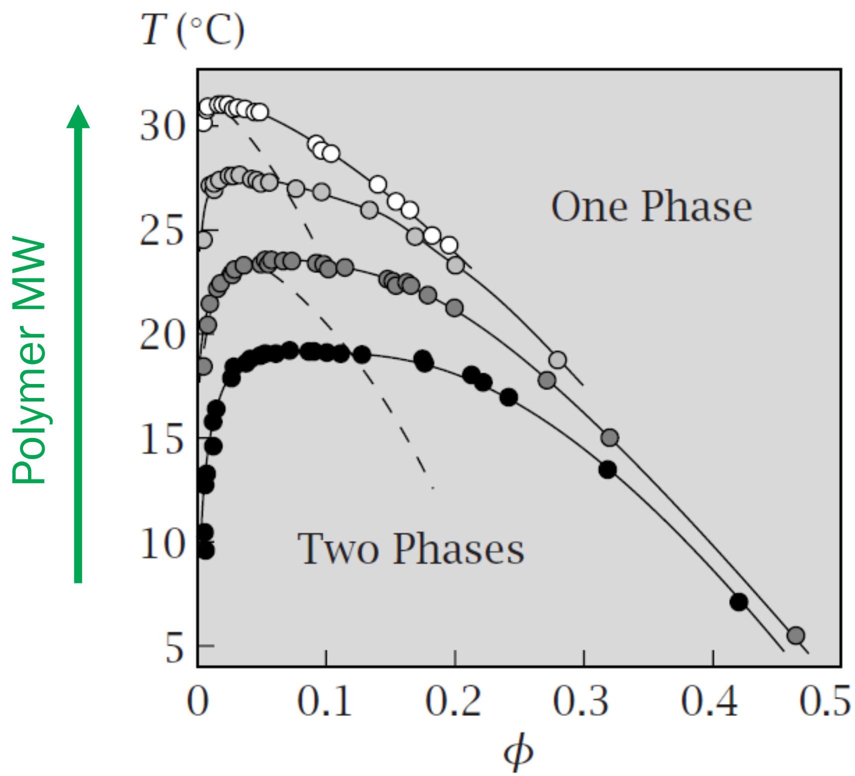
(Image: Dill and Bromberg, ***Molecular Driving Forces***, 2nd Ed., Figures 32.6)

Phase Diagram



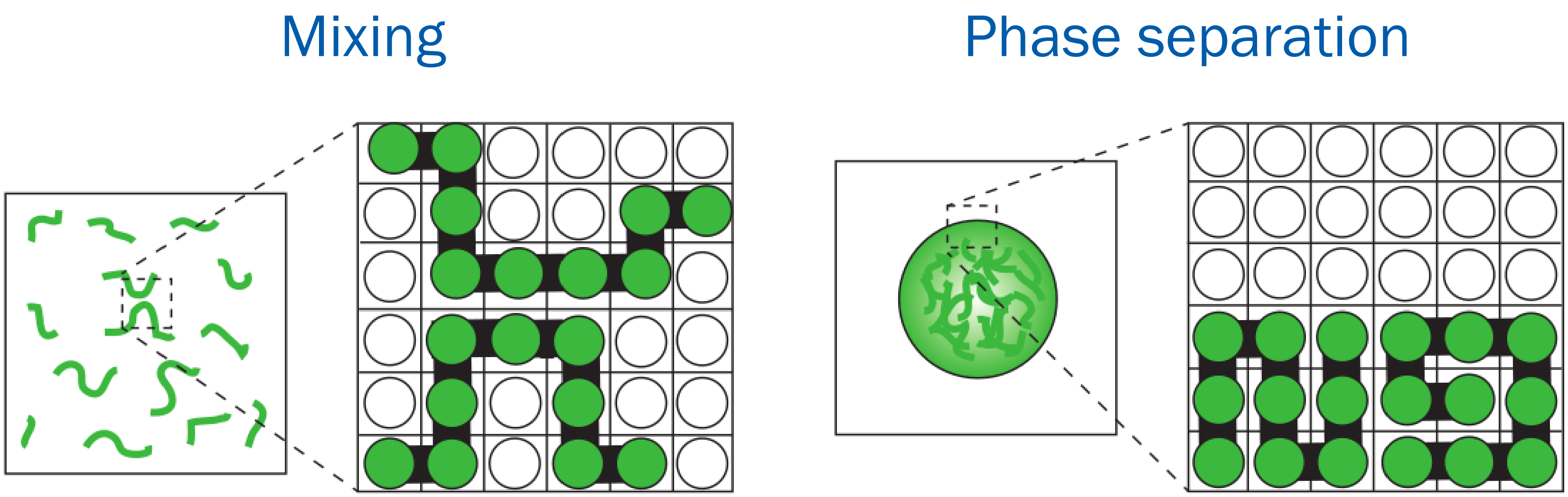
(Image: Dill and Bromberg, *Molecular Driving Forces*, 2nd Ed., Figures 32.7)

Polystyrene in Cyclohexane



(Image: Dill and Bromberg, ***Molecular Driving Forces***, 2nd Ed., Figures 32.8)

Phase Separation of Polymers



Multivalency and flexibility are important

(Image: Brangwynne *et al*., ***Nature Physics*** (2015))

Proteins are Polymer Chains

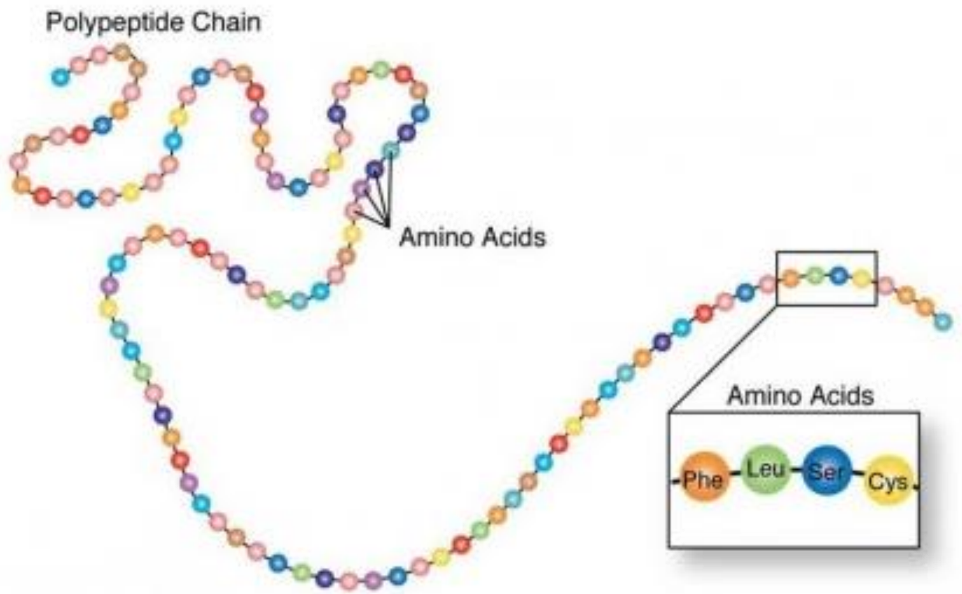
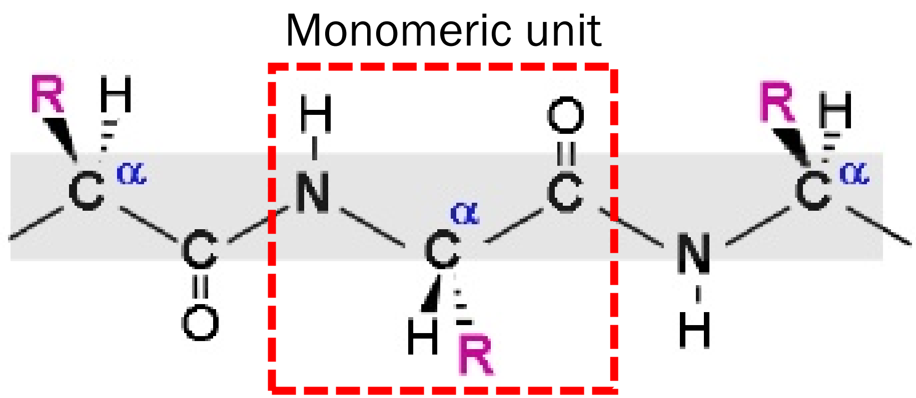


Image: http://www.mcqbiology.com/2012/11/mcq-on-amino-acids.html

Proteins are Polymer Chains



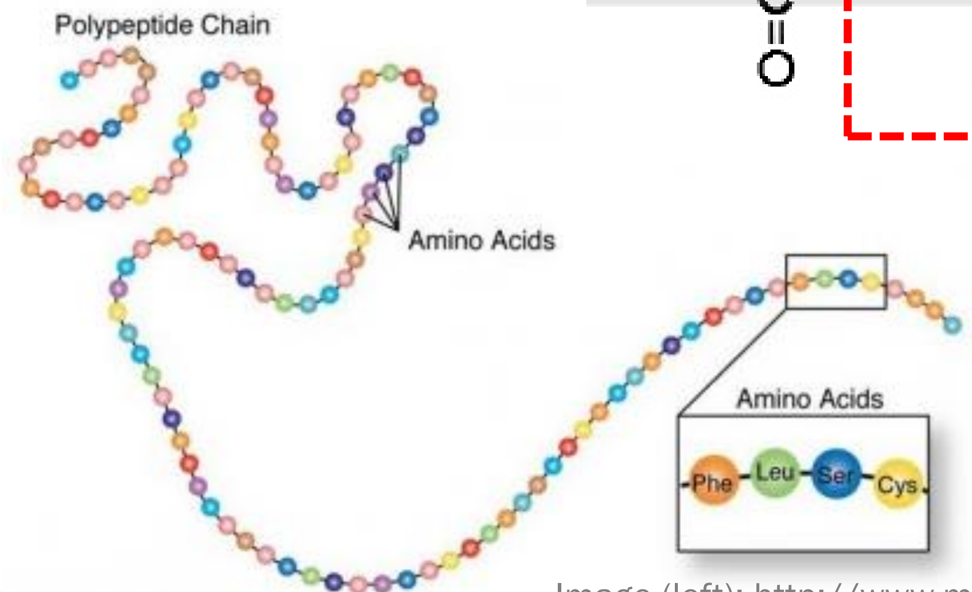
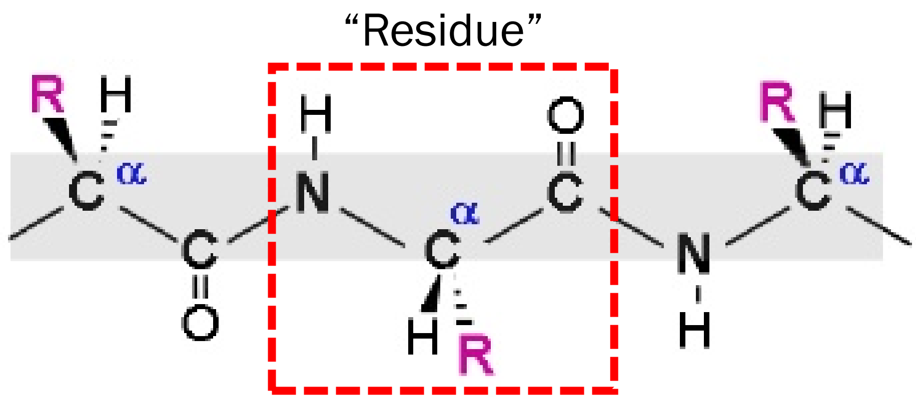
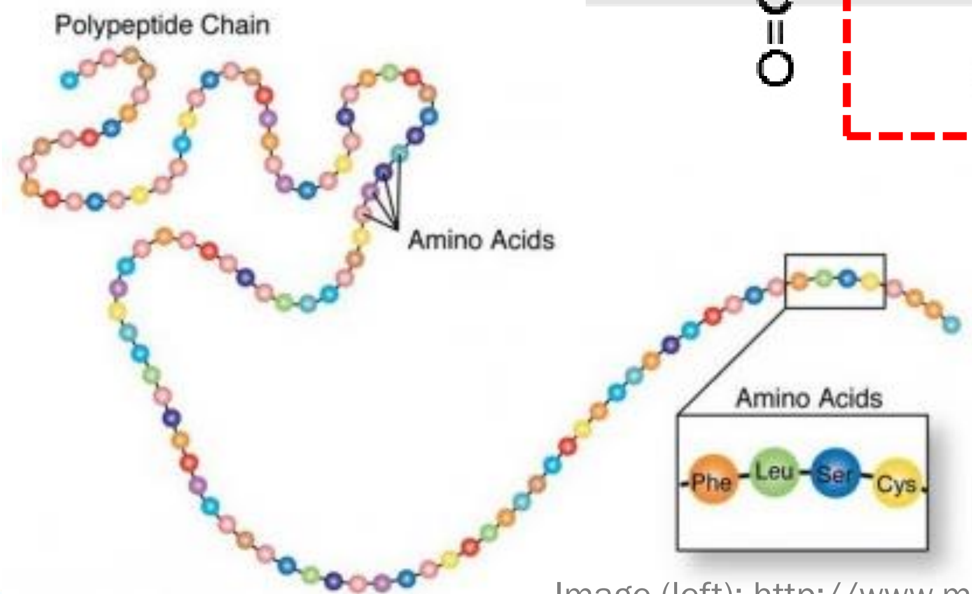


Image (left): http://www.mcqbiology.com/2012/11/mcq-on-amino-acids.html

Image (right): https://www.slideshare.net/MUBOSScz/14-proteins

Proteins are Polymer Chains





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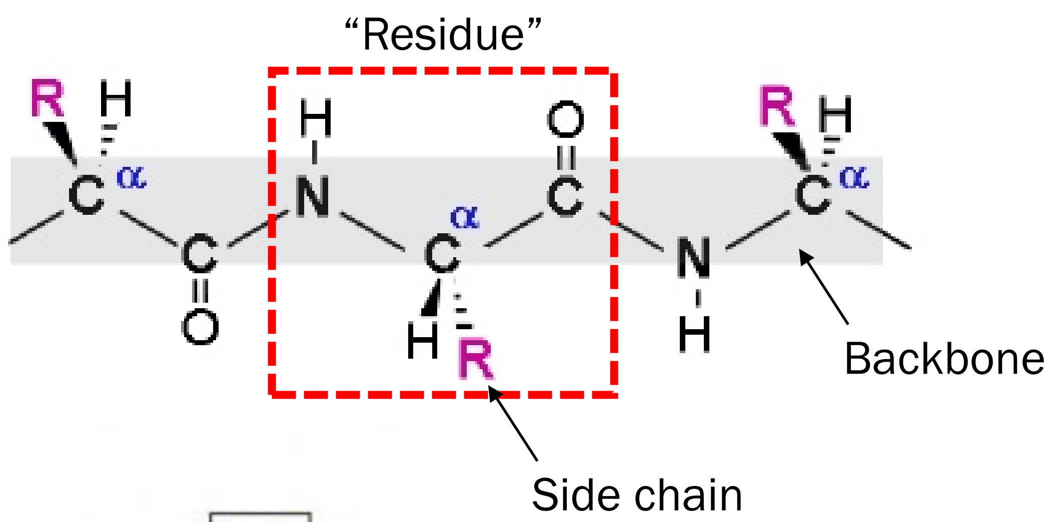

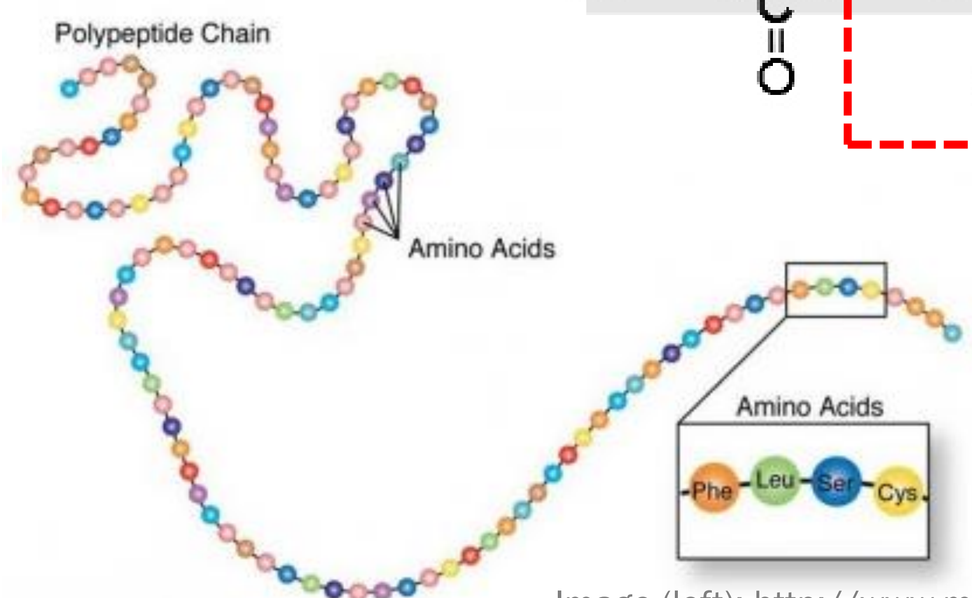

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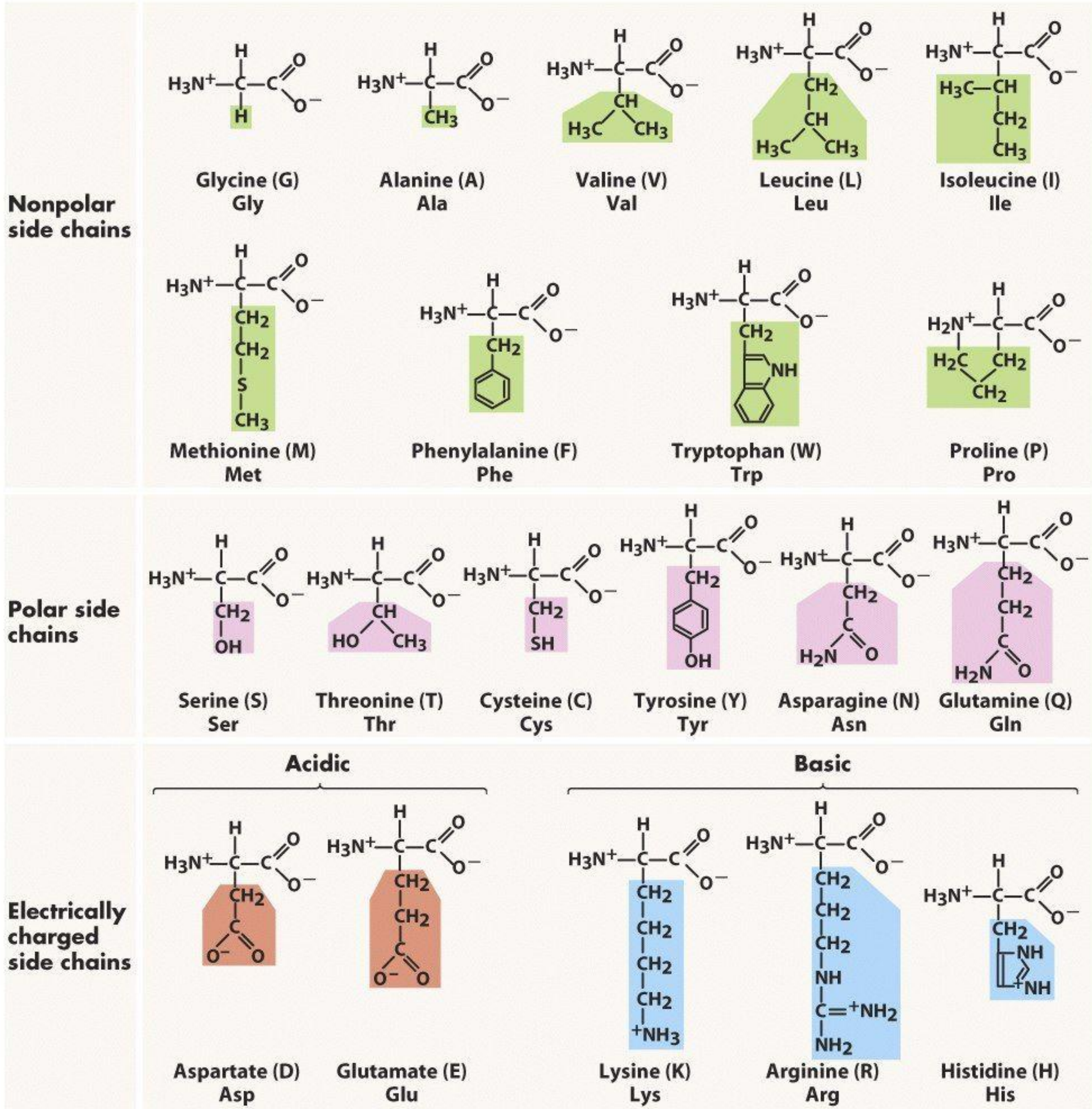


Figure 3-5 Biological Science, 2/e

Canonical Proteins Require Folding

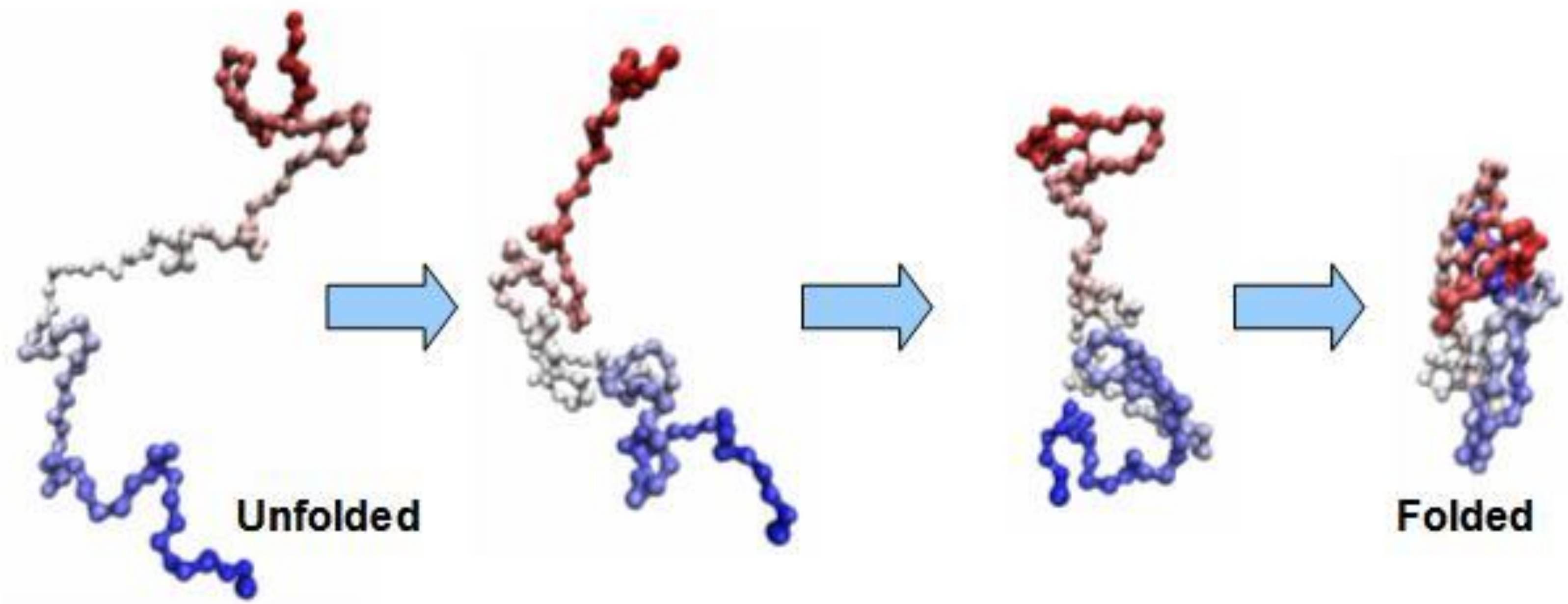
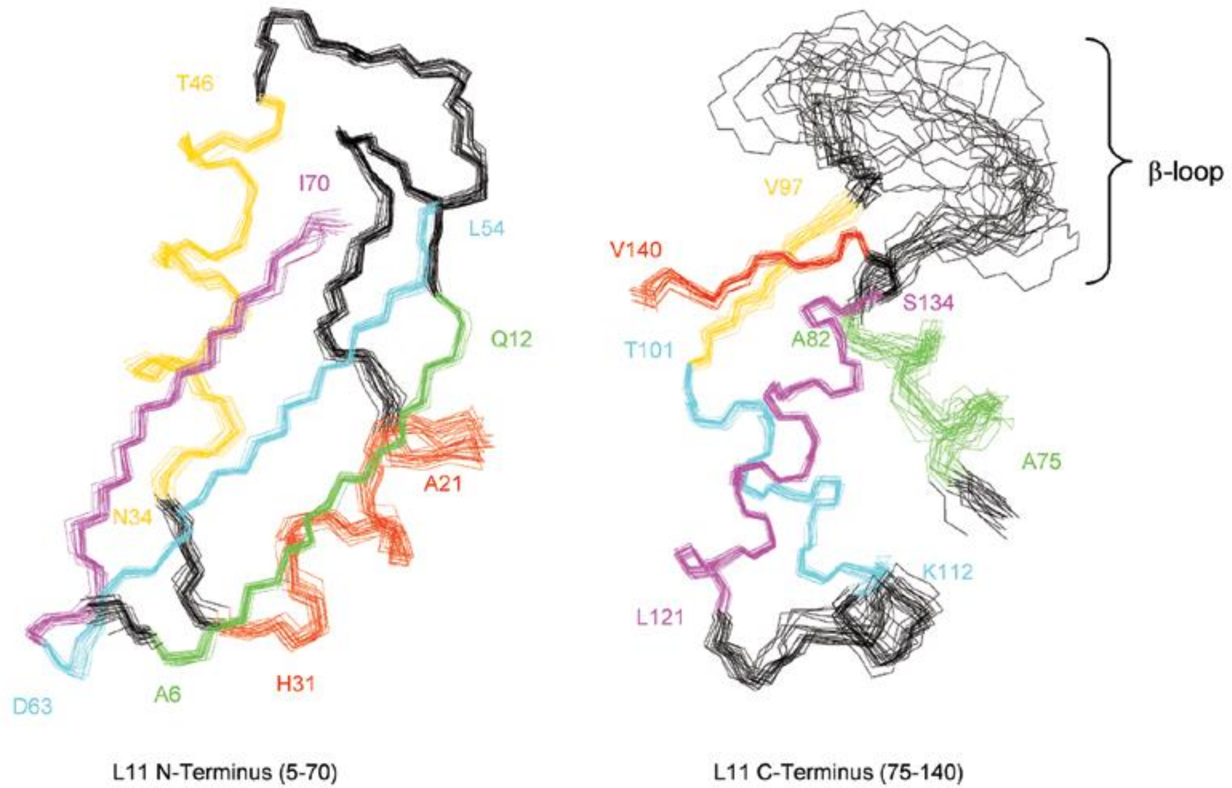


Image: https://inst.eecs.berkeley.edu/~ee127a/book/login/exa_protein_folding.html

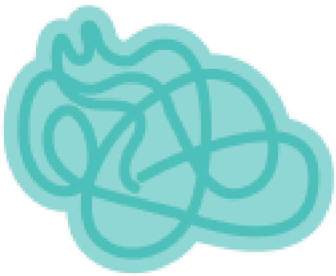
“Frozen” in the Folded State



(measurement by NMR)

Image: http://schwalbe.org.chemie.uni-frankfurt.de/node/684

Thermodynamics of Folding



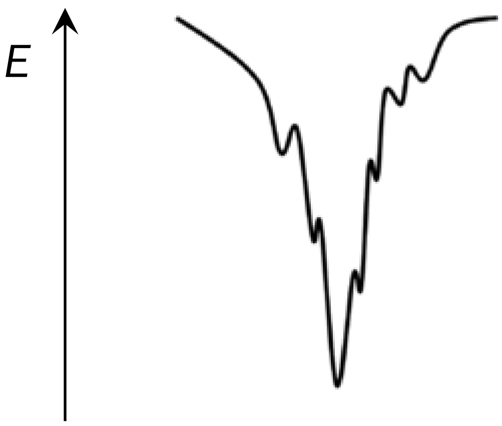
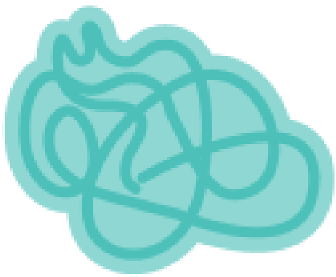
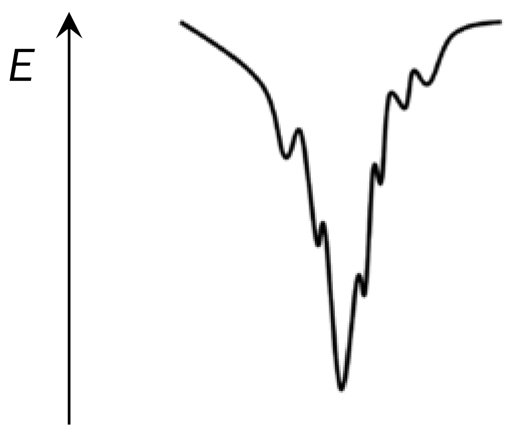


Image (top): https://www.quantamagazine.org/how-disordered-proteins-are-upending-molecular-biology-20170118

Image (bottom): Mészáros *et al.*, ***Phys Biol*** (2011)

Thermodynamics of Folding





Limited flexibility

↓

Almost no chance of phase separation

Image (top): https://www.quantamagazine.org/how-disordered-proteins-are-upending-molecular-biology-20170118

Image (bottom): Mészáros *et al.*, ***Phys Biol*** (2011)

What If …

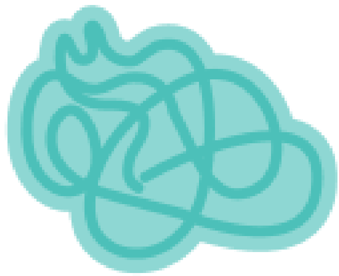


Image (top): https://www.quantamagazine.org/how-disordered-proteins-are-upending-molecular-biology-20170118

Image (bottom): Mészáros *et al*., ***Phys Biol*** (2011)

No Fixed Structure: Flexibility!

Image (top): https://www.quantamagazine.org/how-disordered-proteins-are-upending-molecular-biology-20170118

Image (bottom): Mészáros *et al*., ***Phys Biol*** (2011)

Intrinsically Disordered Protein (IDP)

foldable protein intrinsically disordered protein

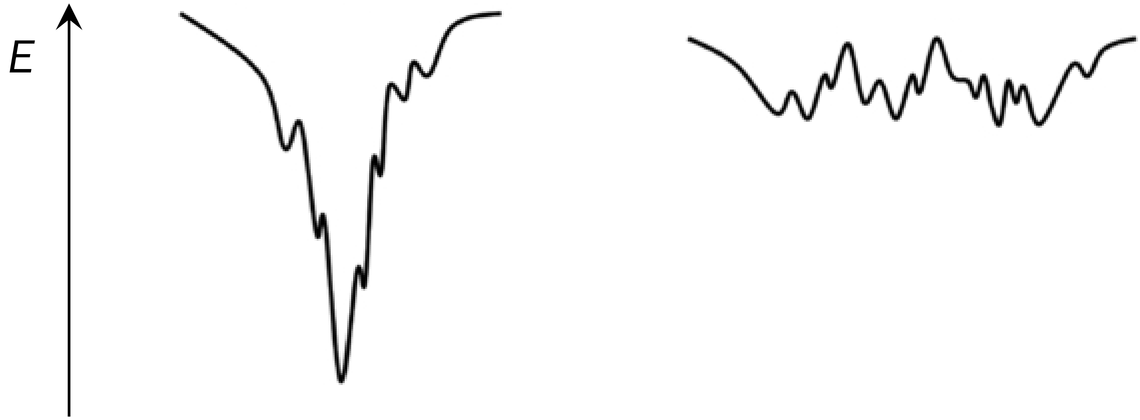
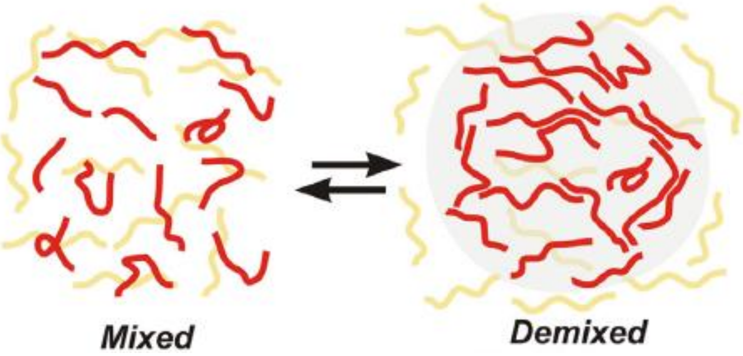
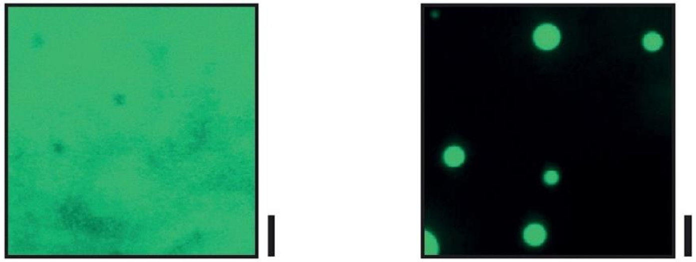


Image (top): https://www.quantamagazine.org/how-disordered-proteins-are-upending-molecular-biology-20170118

Image (bottom): Mészáros *et al.*, ***Phys Biol*** (2011)

IDPs and Phase Separation

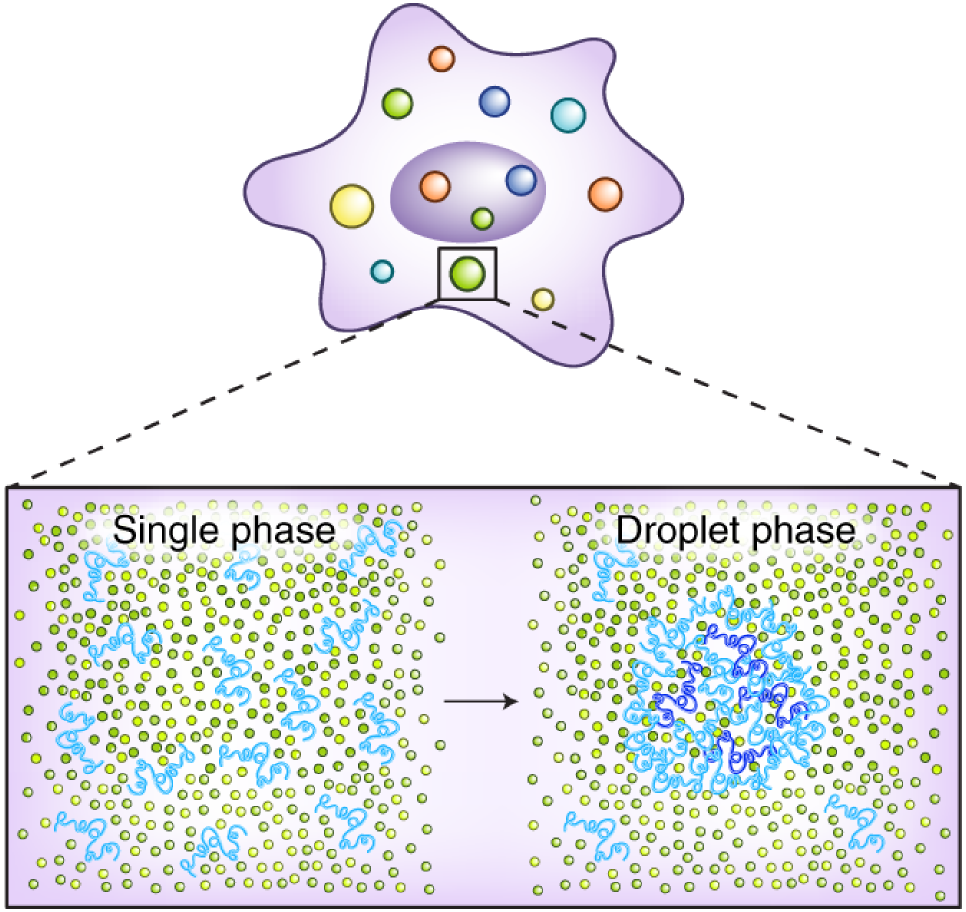




(Scale bars: 5 μm)

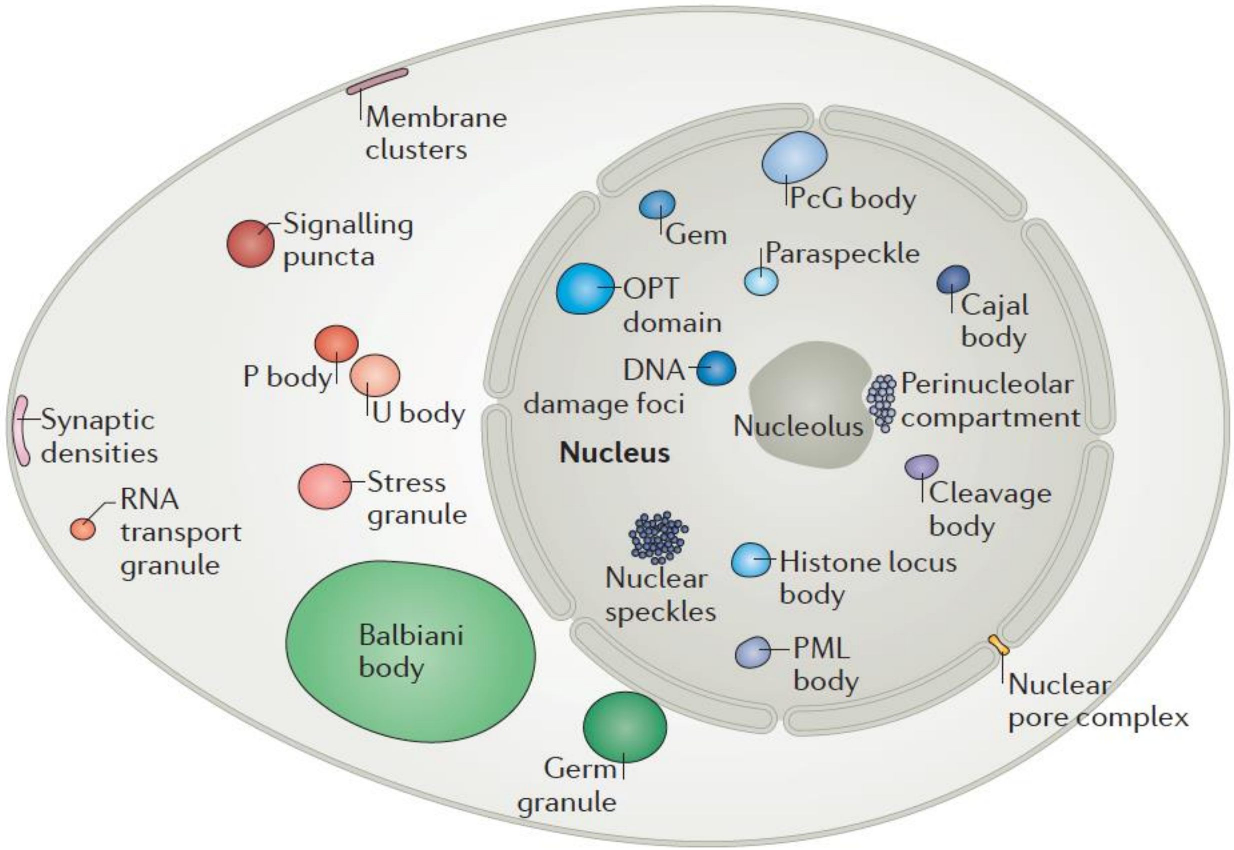
Alberti and Hyman, ***BioEssays*** (2016); Boeynaems *et al*., ***Trends Cell Biol*** (2018)

Intracellular Phase Separation



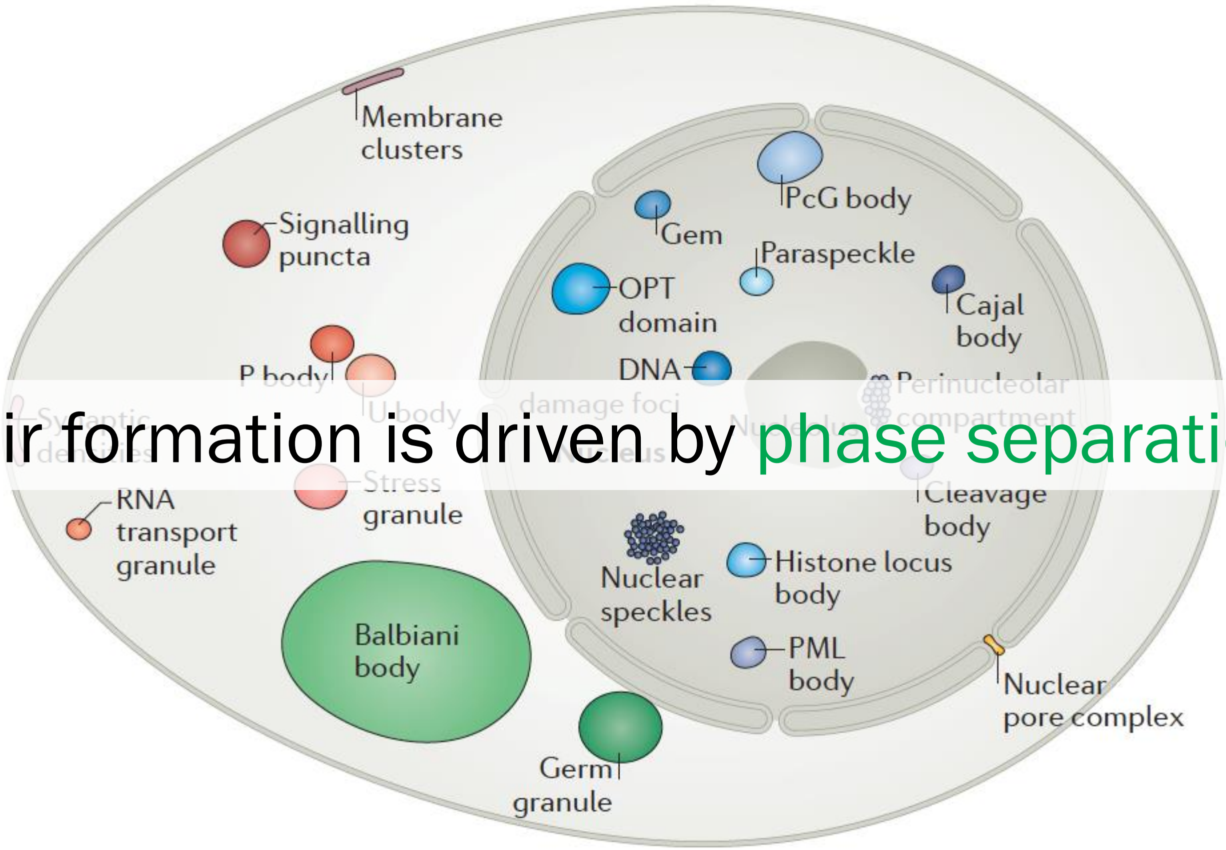
Tang, *Nature Methods* (2019)

Membraneless Organelles (MLOs)



Banani *et al.*, ***Nat Rev Cell Biol*** (2017)

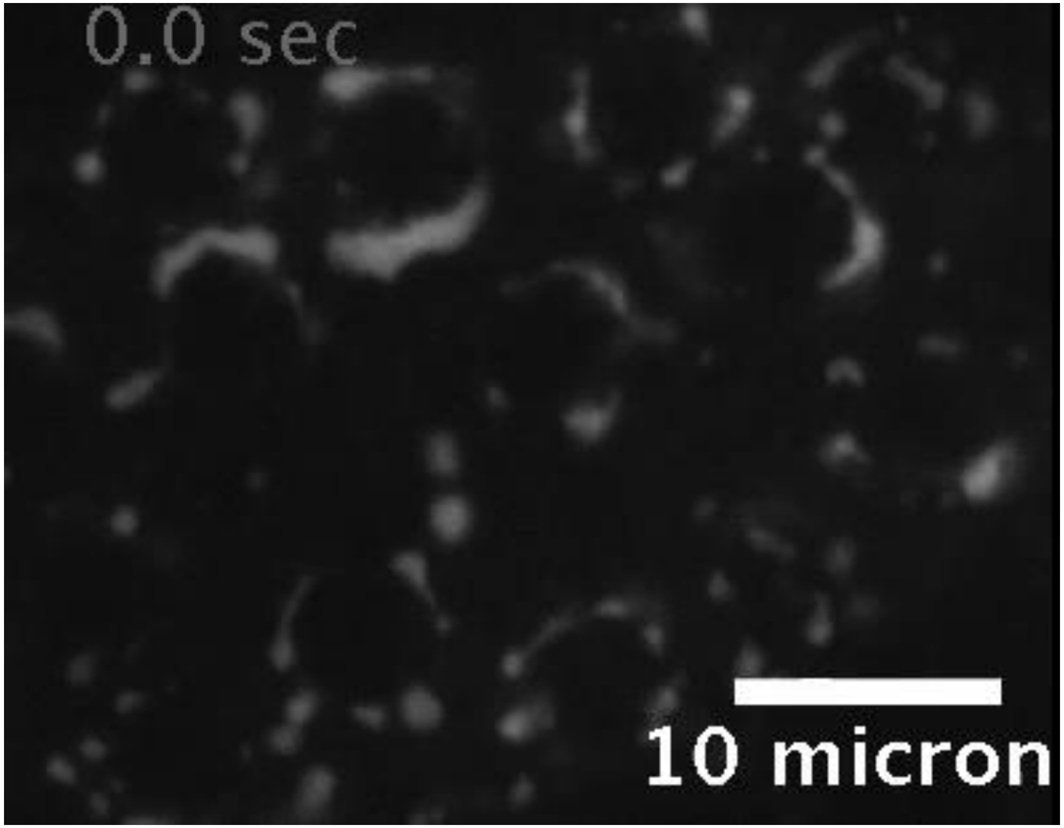
Membraneless Organelles (MLOs)



Their formation is driven by phase separation!

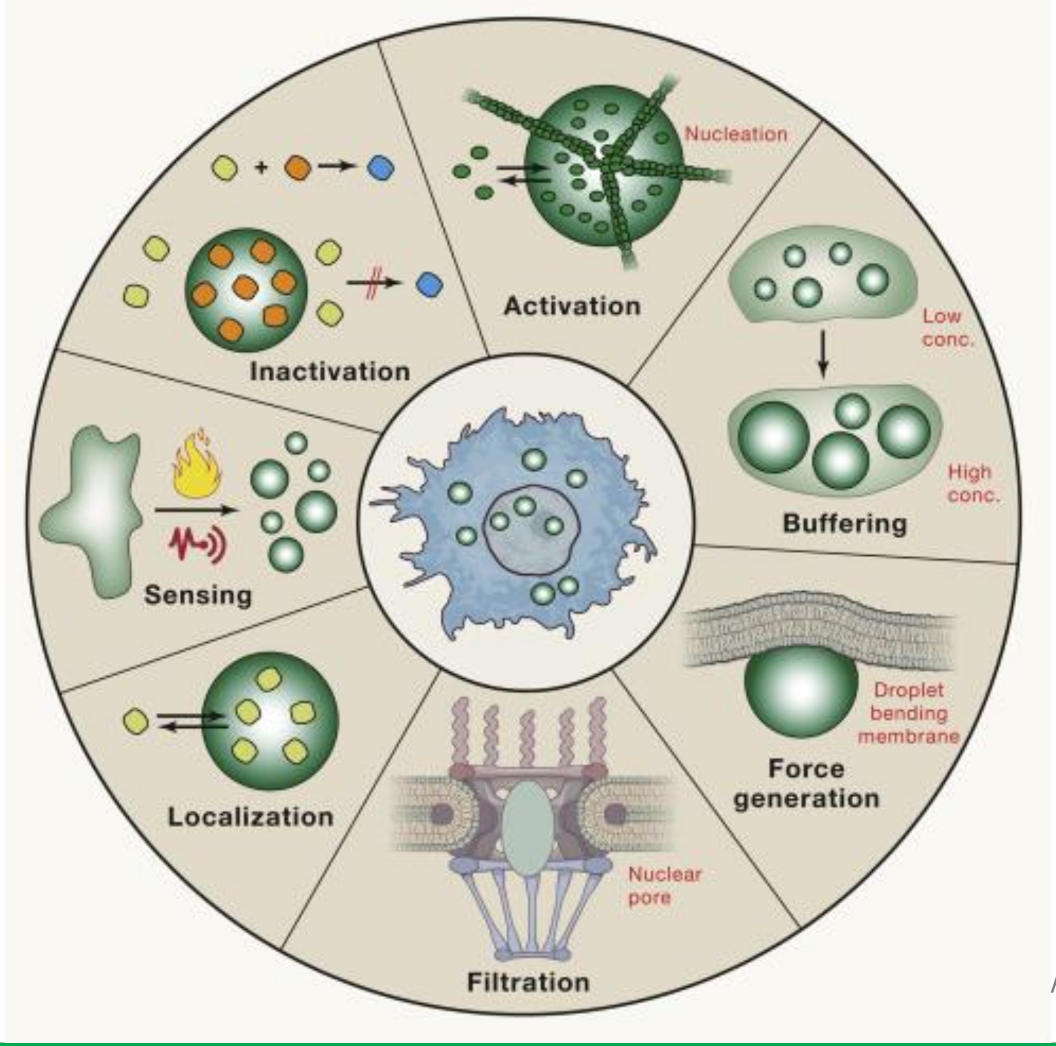
Banani *et al.*, ***Nat Rev Cell Biol*** (2017)

Liquidity of MLOs (even *in vivo*)



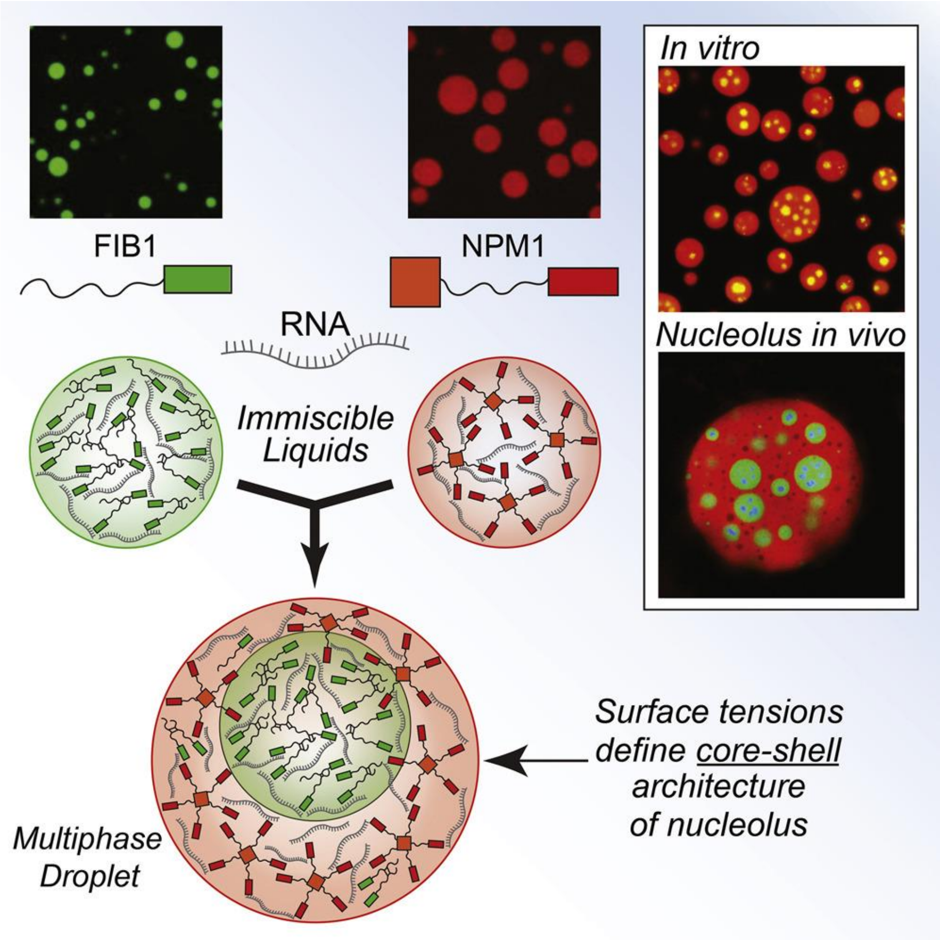
Brangwynne *et al.*, ***Science*** (2009)

Biological Roles of MLOs



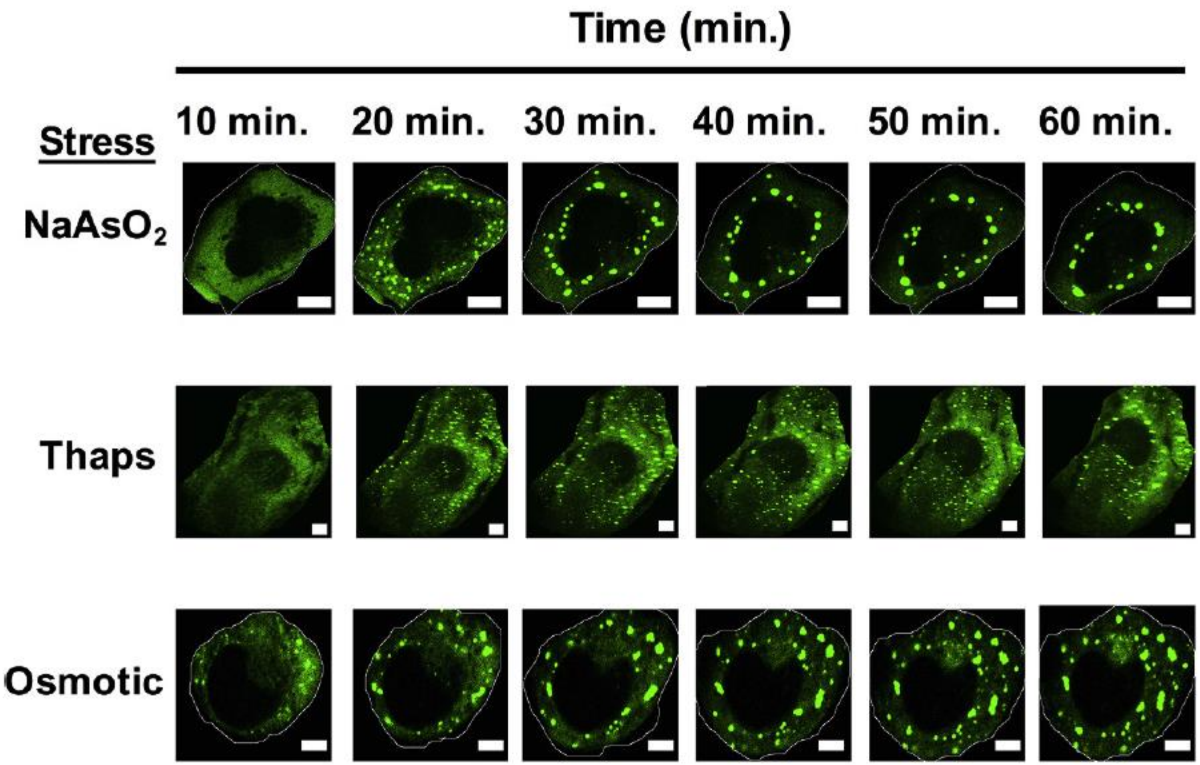
Alberti *et al*., *Cell* (2019)

Example: Nucleolus Formation



Feric *et al*., ***Cell*** (2016)

Example: Stress Granules



(Bar: 5 μm)

human osteosarcoma cells (U-2 OS)
Signal from GFP-G3BP1 fusion protein

Wheeler *et al.*, ***eLife*** (2016)

LLPS and Neuroscience

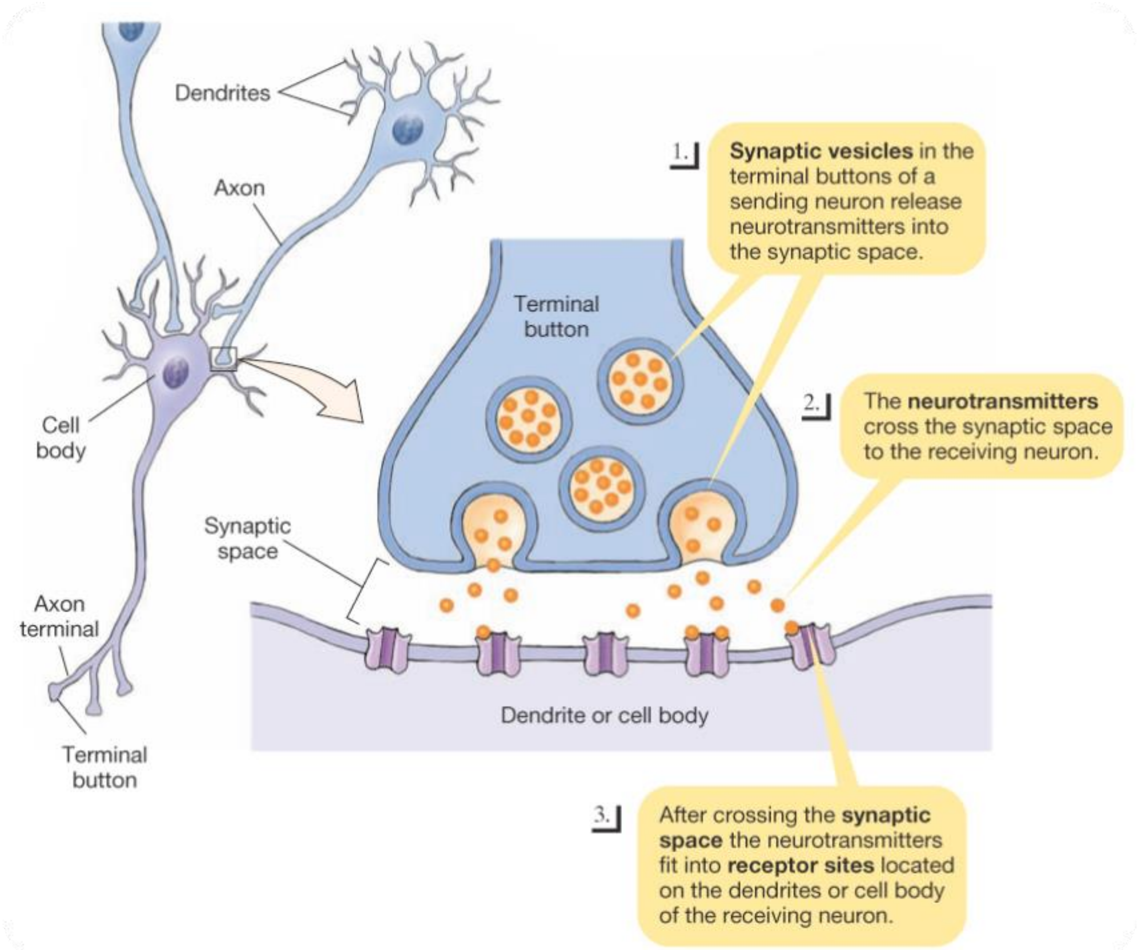
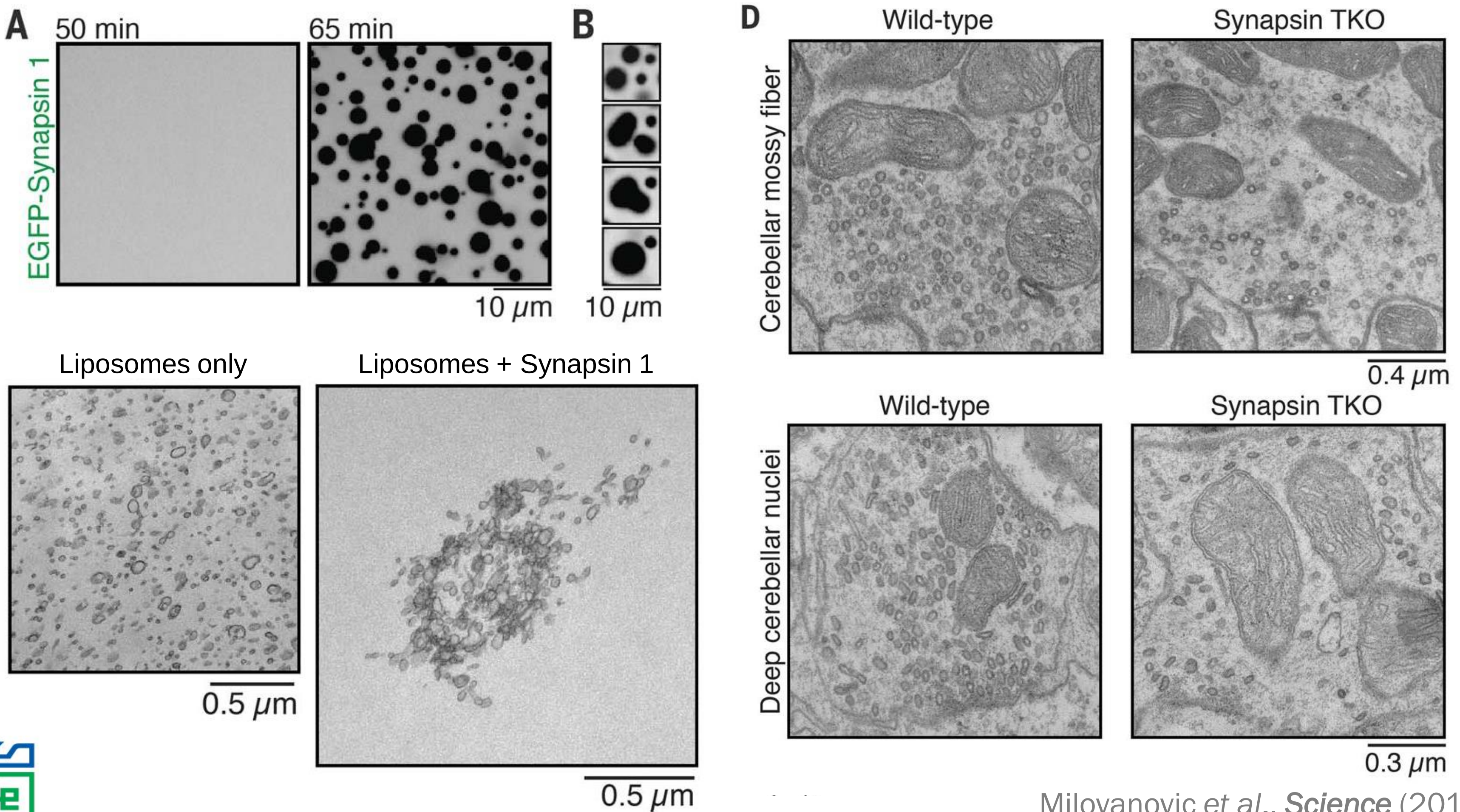


Image: https://socratic.org/questions/how-do-impulses-travel-across-a-synapse

LLPS and Neuroscience

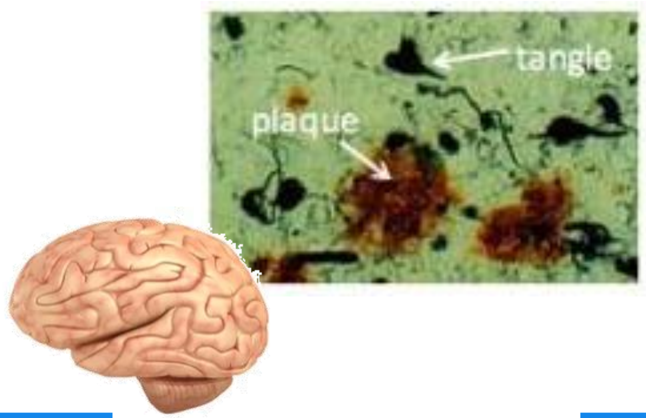


Milovanovic *et al.*, ***Science*** (2018)

Neurodegenerative Diseases

Alzheimer's disease
amyloid beta, tau

Parkinson's disease
α-synuclein



Huntington's disease
huntingtin

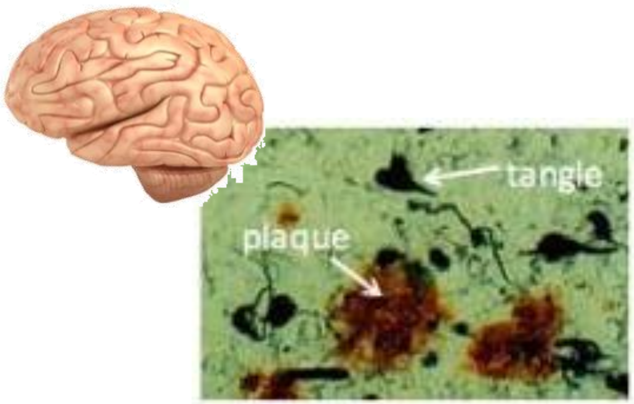
BSE (mad cow disease)
prion

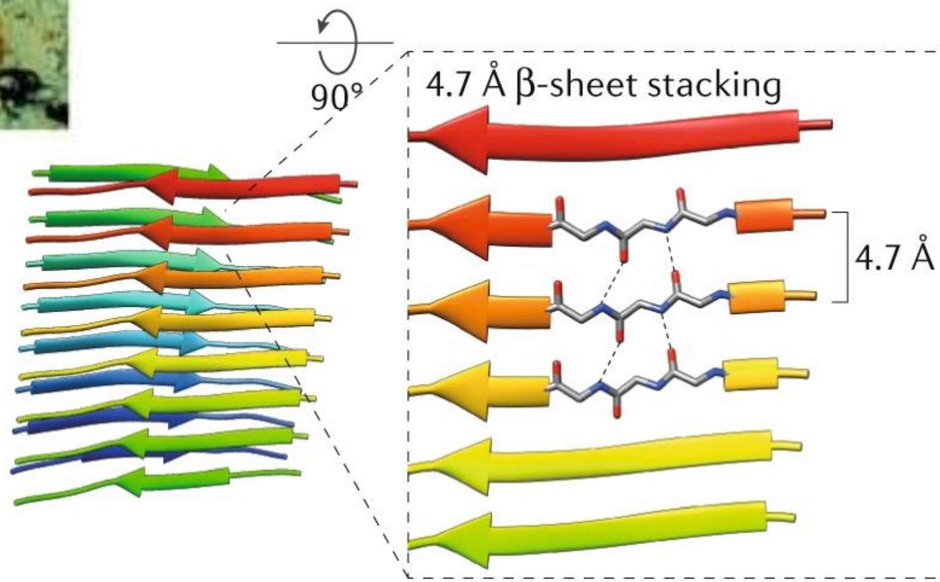
ALS (Lou Gehrig's disease)
TDP43, FUS

Images: https://www.livescience.com/29365-human-brain.html

http://www.chemistry.emory.edu/faculty/lynn/Yan_Liang.html

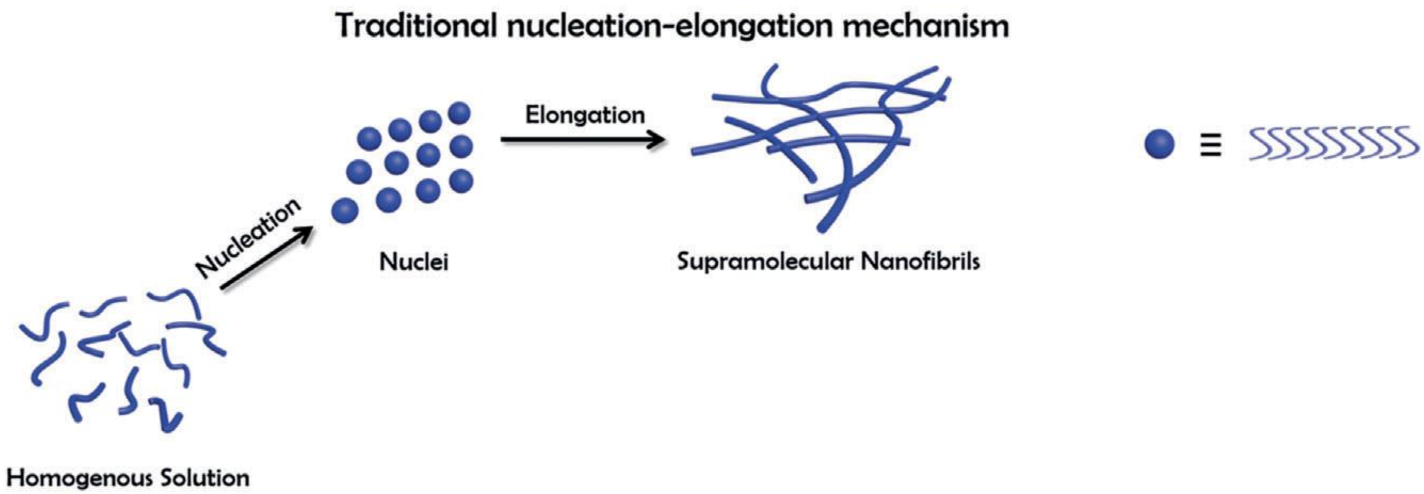
Deposition of Amyloid Fibrils





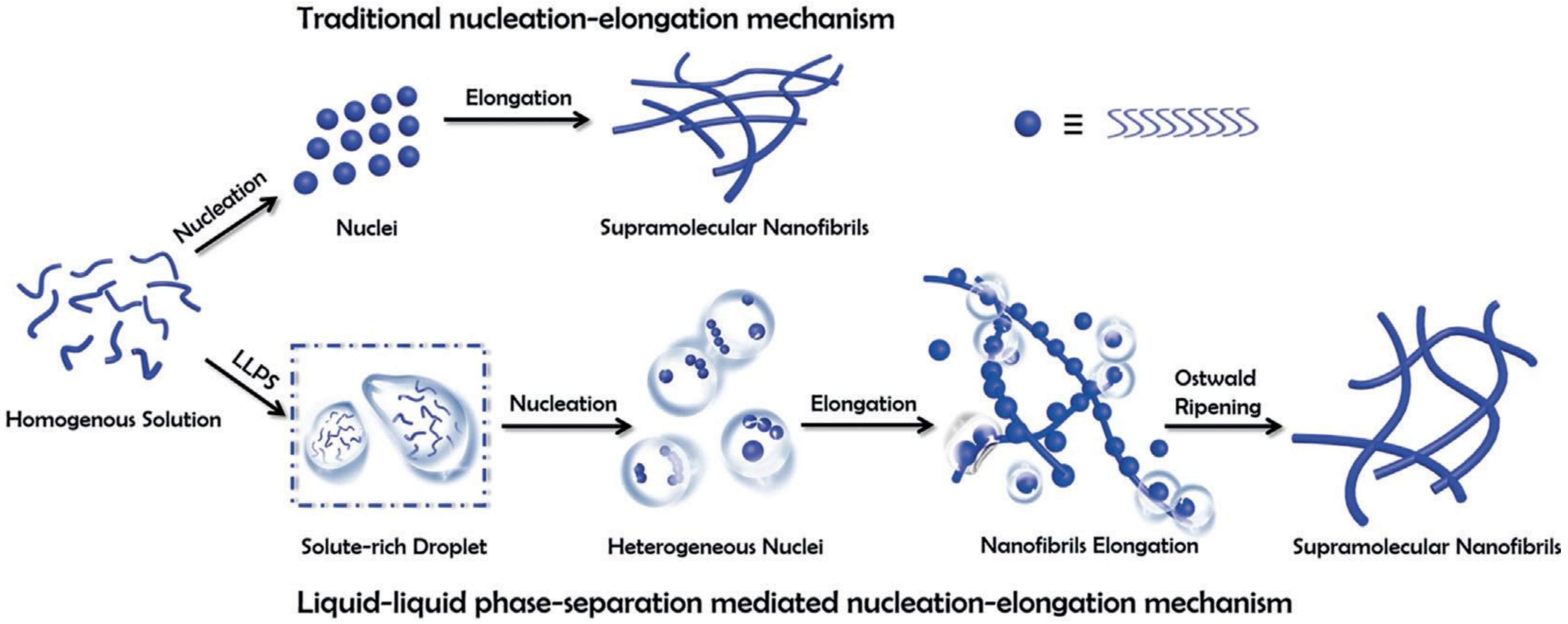
Iadanza *et al.*, ***Nat Rev Mol Cell Biol*** (2018)

Traditional Hypothesis



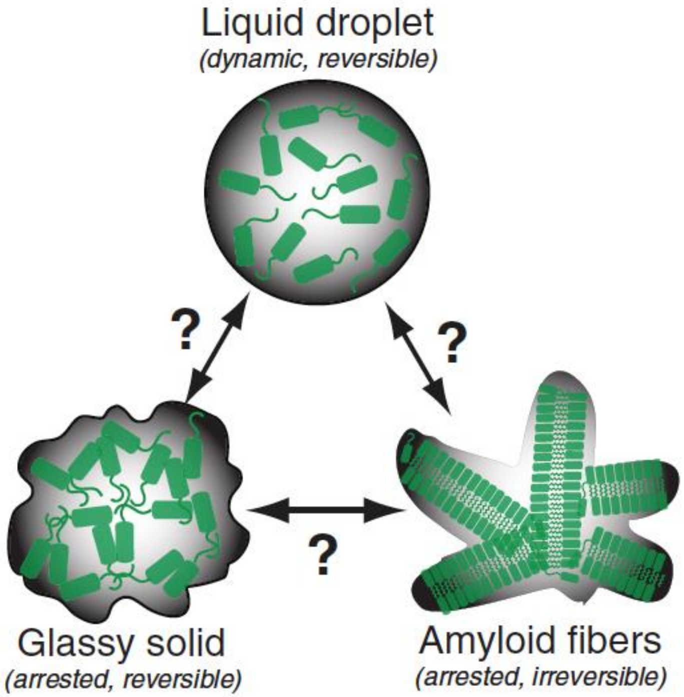
Yuan *et al.*, *Angew Chem Int Ed* (2019)

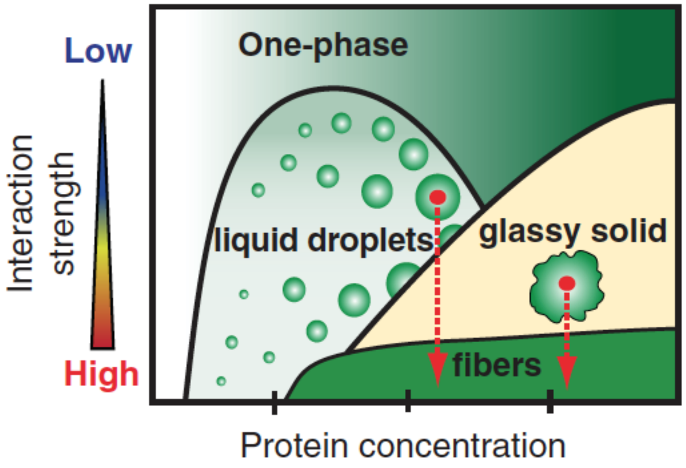
Another Route: via Phase Separation



Yuan *et al.*, ***Angew Chem Int Ed*** (2019)

IDPs Show Rich Phase Behaviors



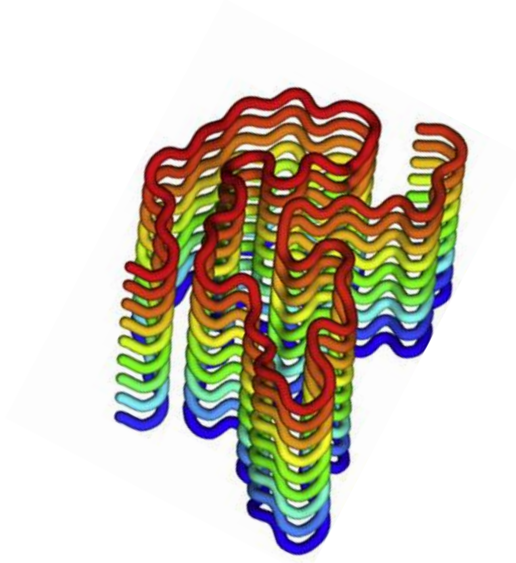


Shin and Brangwynne, *Science* (2017)

FUS (*Fused in Sarcoma*) Protein

DNA/RNA binding

Fibril formation in ALS



Stress response

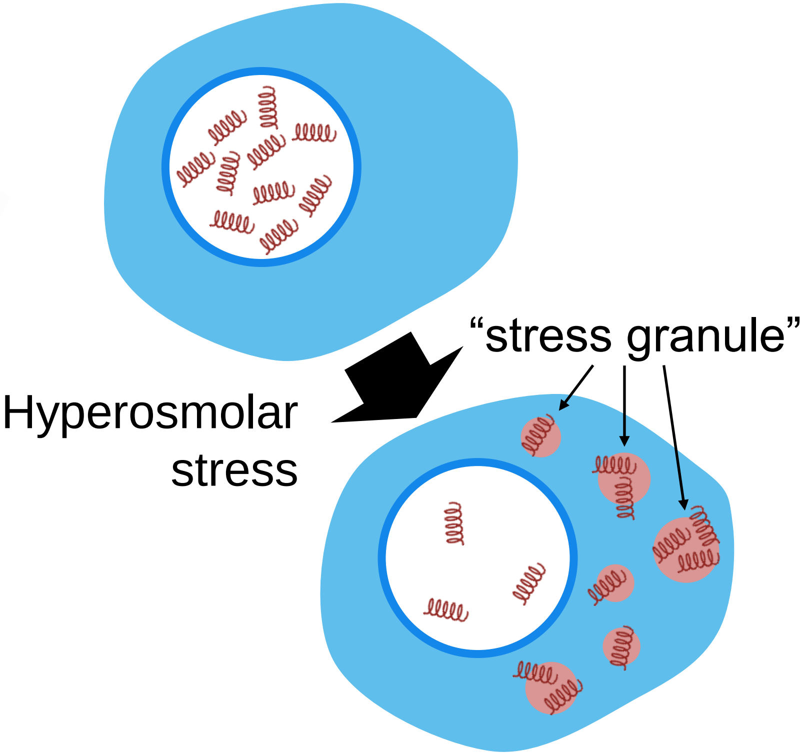
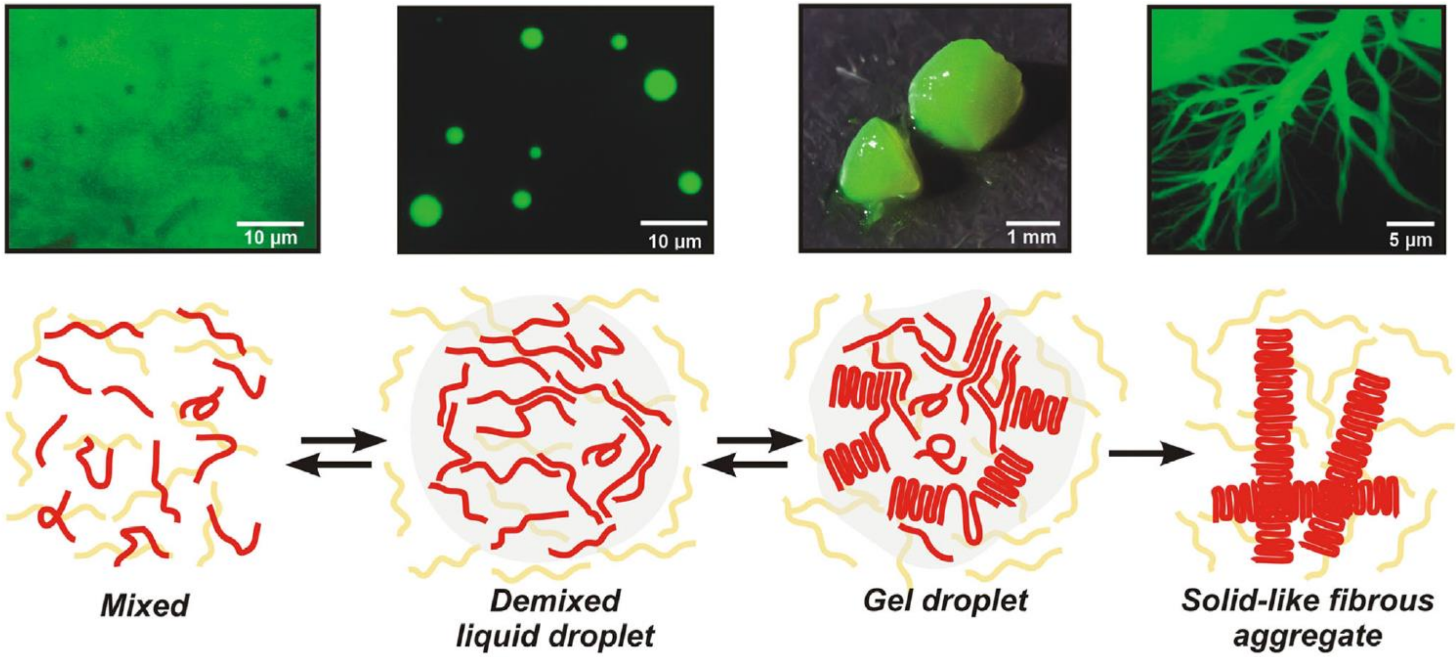


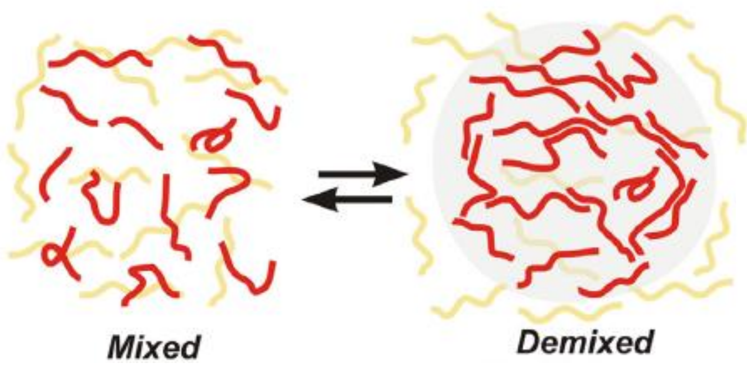
Image: https://www.vecteezy.com/vector-art/88822-dna-rna-vectors; Sama *et al*., ***ASN Neuro*** (2014); Murray *et al*., ***Cell*** (2017)

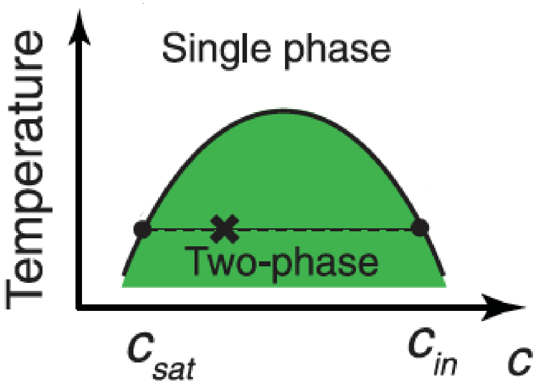
Phase Behaviors of FUS



Alberti and Hyman, *BioEssays* (2016)

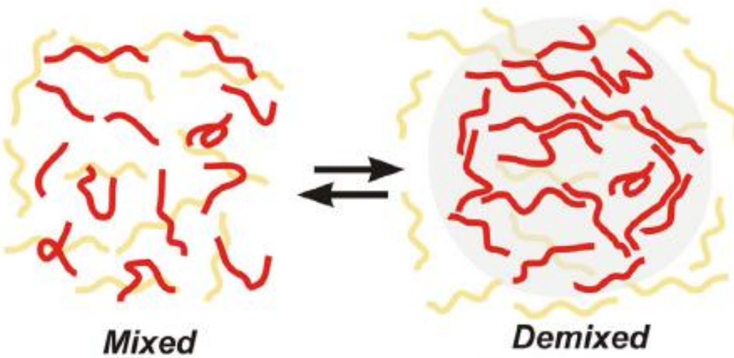
Readout for Phase Behavior

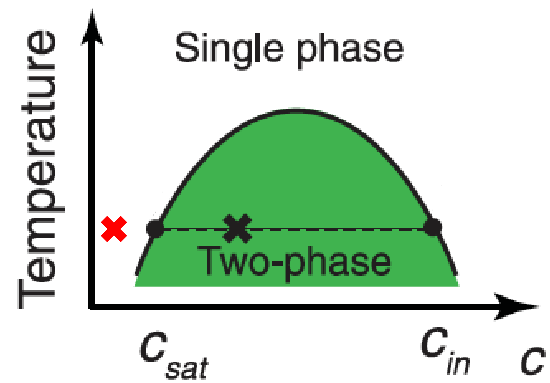




Shin and Brangwynne, ***Science*** (2017); Alberti and Hyman, ***BioEssays*** (2016)

Readout for Phase Behavior

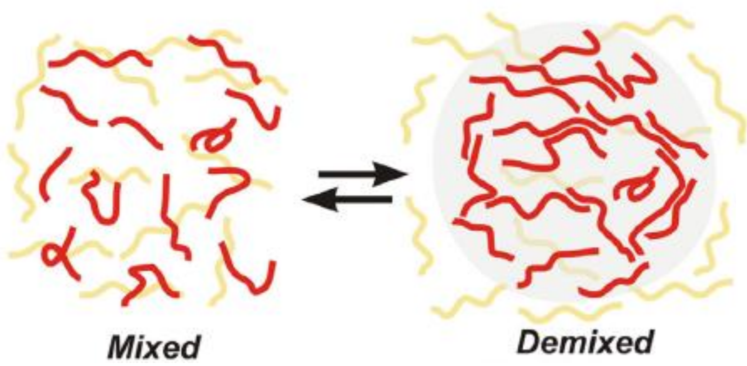


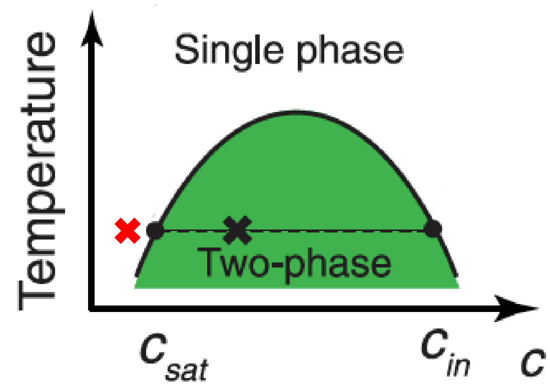


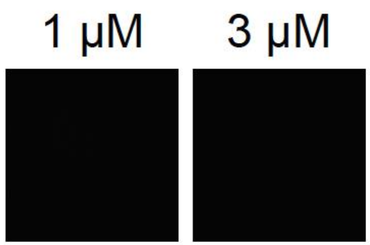
1 µM

Wang *et al*., ***Cell*** (2018)

Readout for Phase Behavior

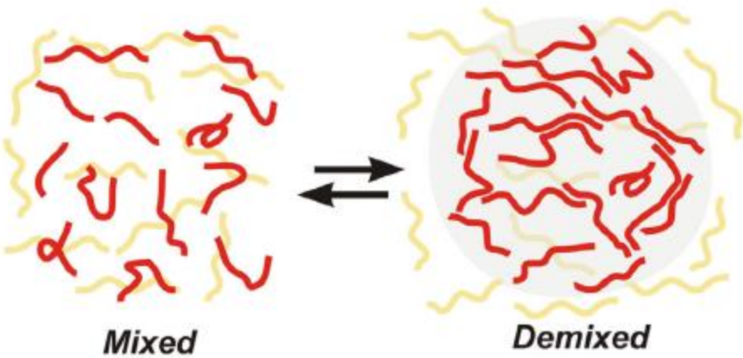


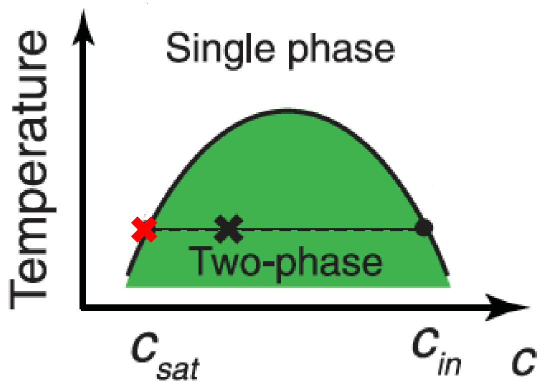


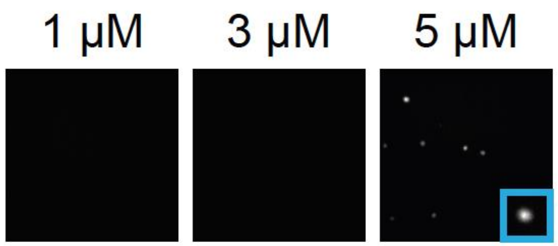


Wang *et al*., ***Cell*** (2018)

Readout for Phase Behavior

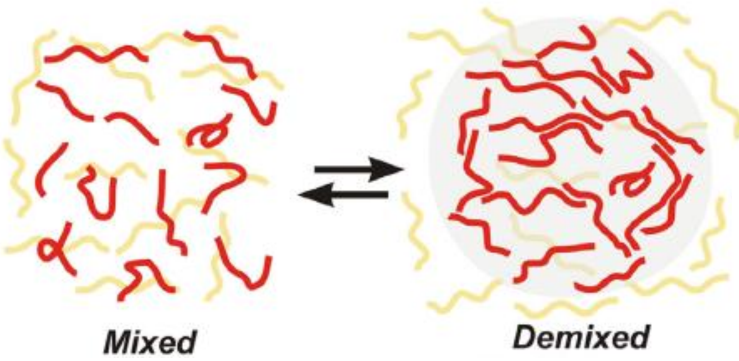


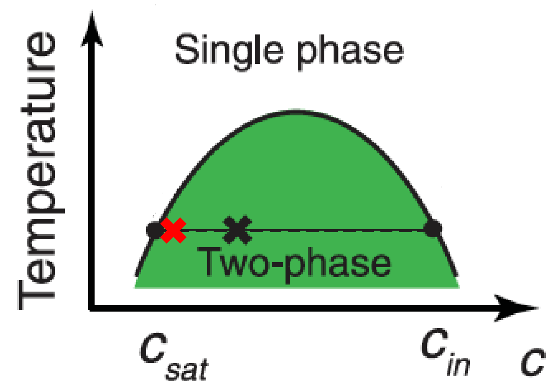


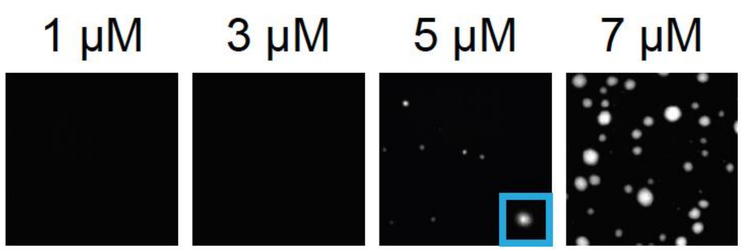


Wang *et al*., ***Cell*** (2018)

Readout for Phase Behavior

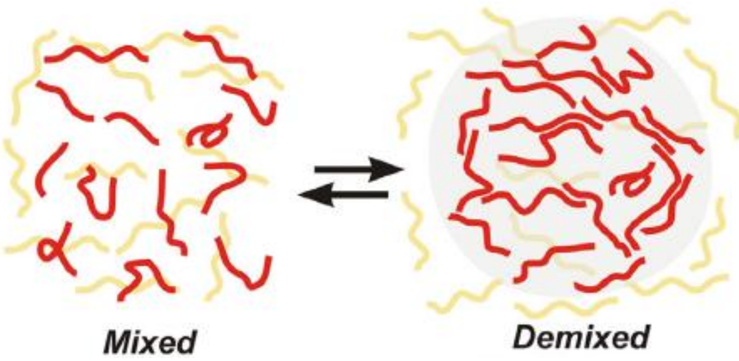


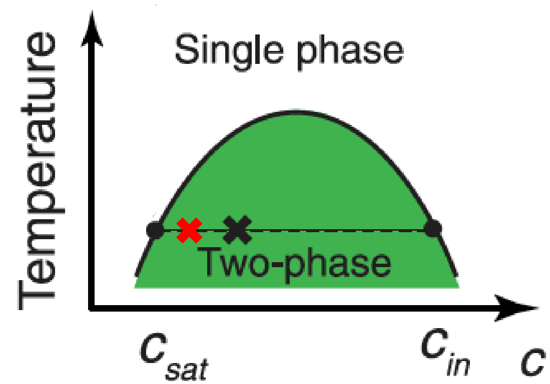


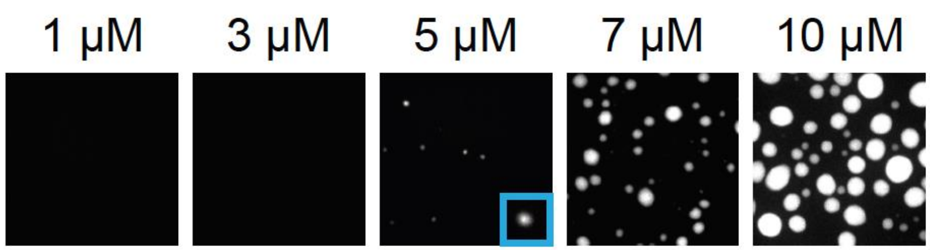


Wang *et al*., ***Cell*** (2018)

Readout for Phase Behavior

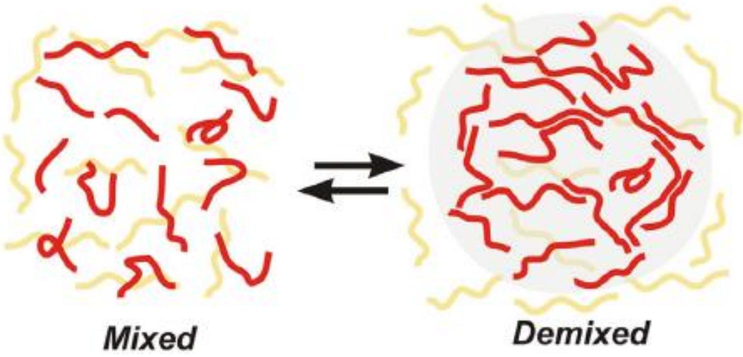


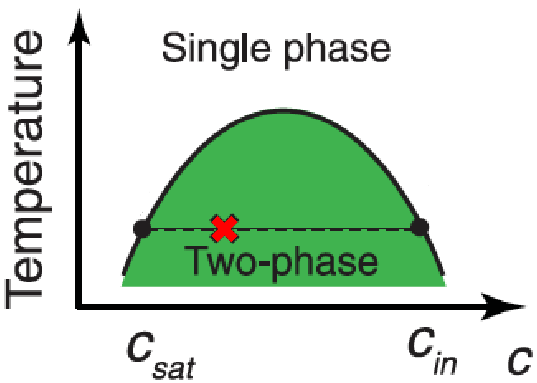


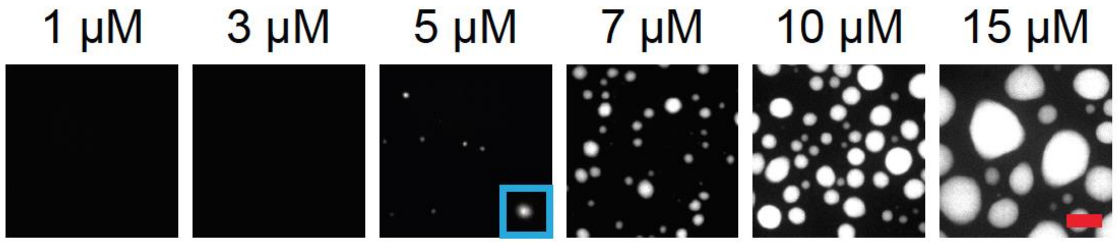


Wang *et al*., ***Cell*** (2018)

Readout for Phase Behavior



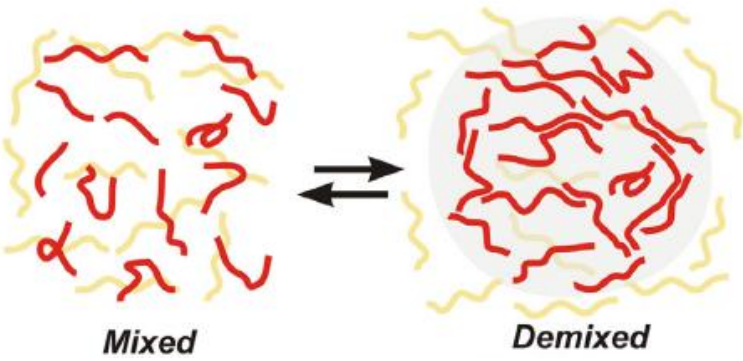


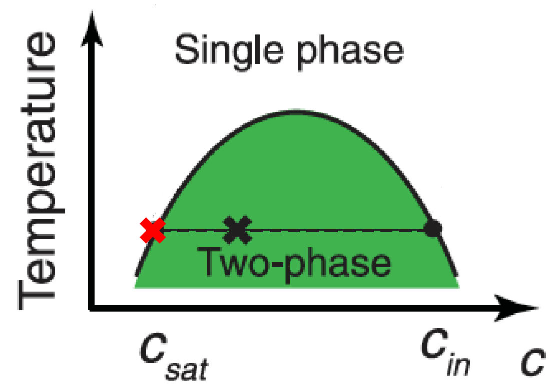


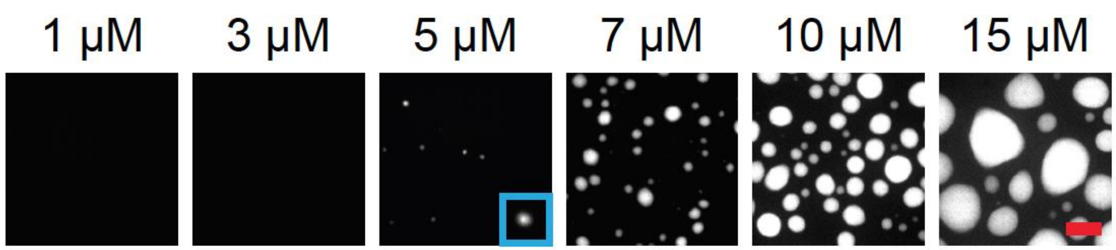
Bar: 5 μm

Wang *et al*., ***Cell*** (2018)

Readout for Phase Behavior





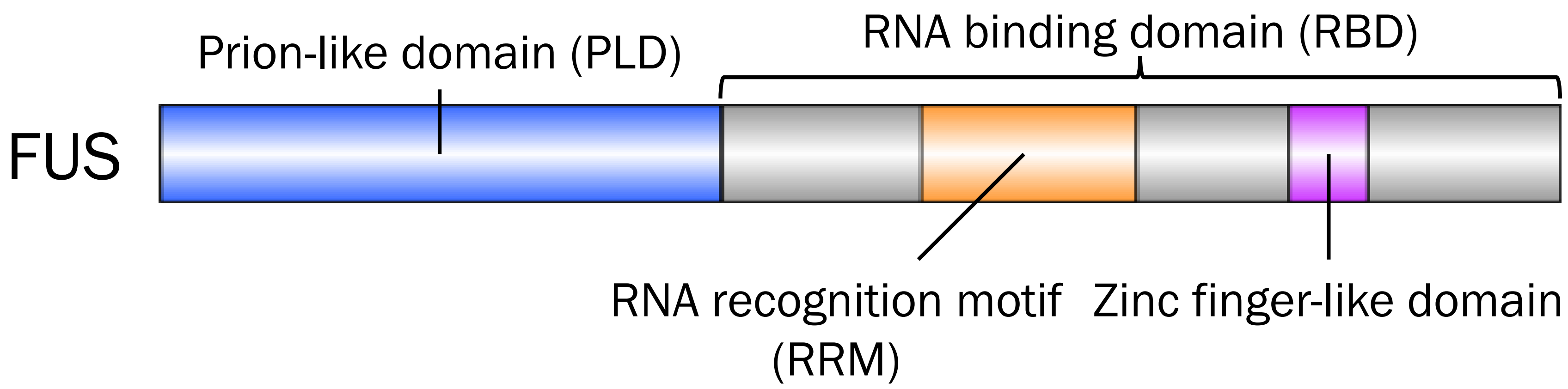


Bar: 5 μm

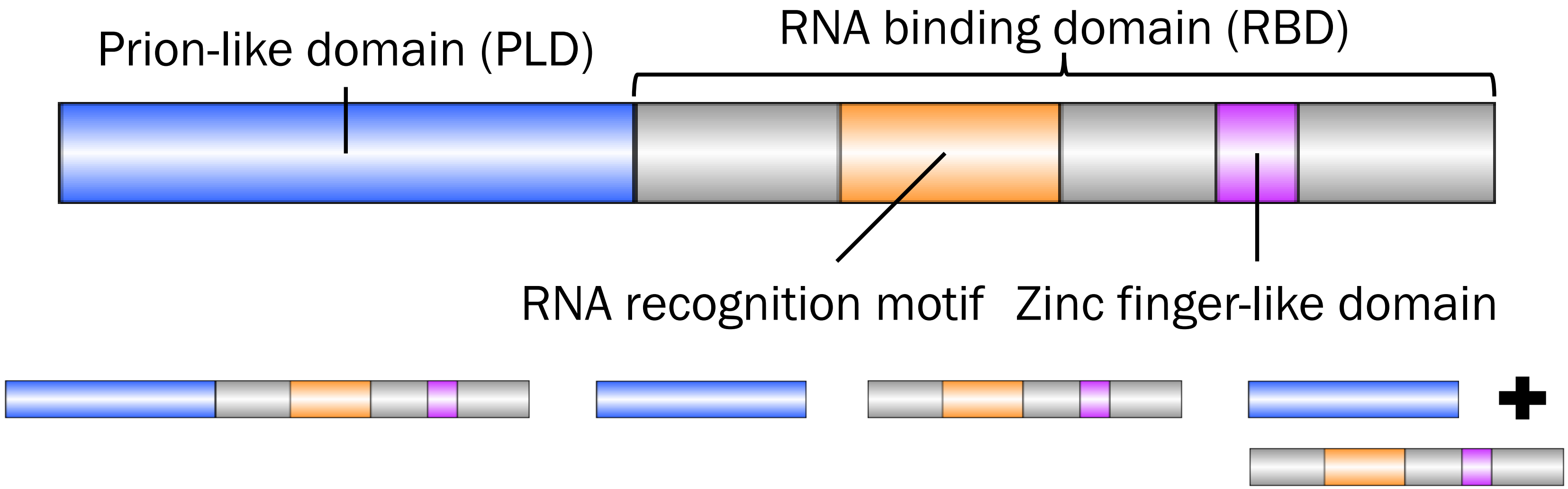
saturation concentration c_{sat}

Wang *et al*., ***Cell*** (2018)

Domain Architecture of FUS

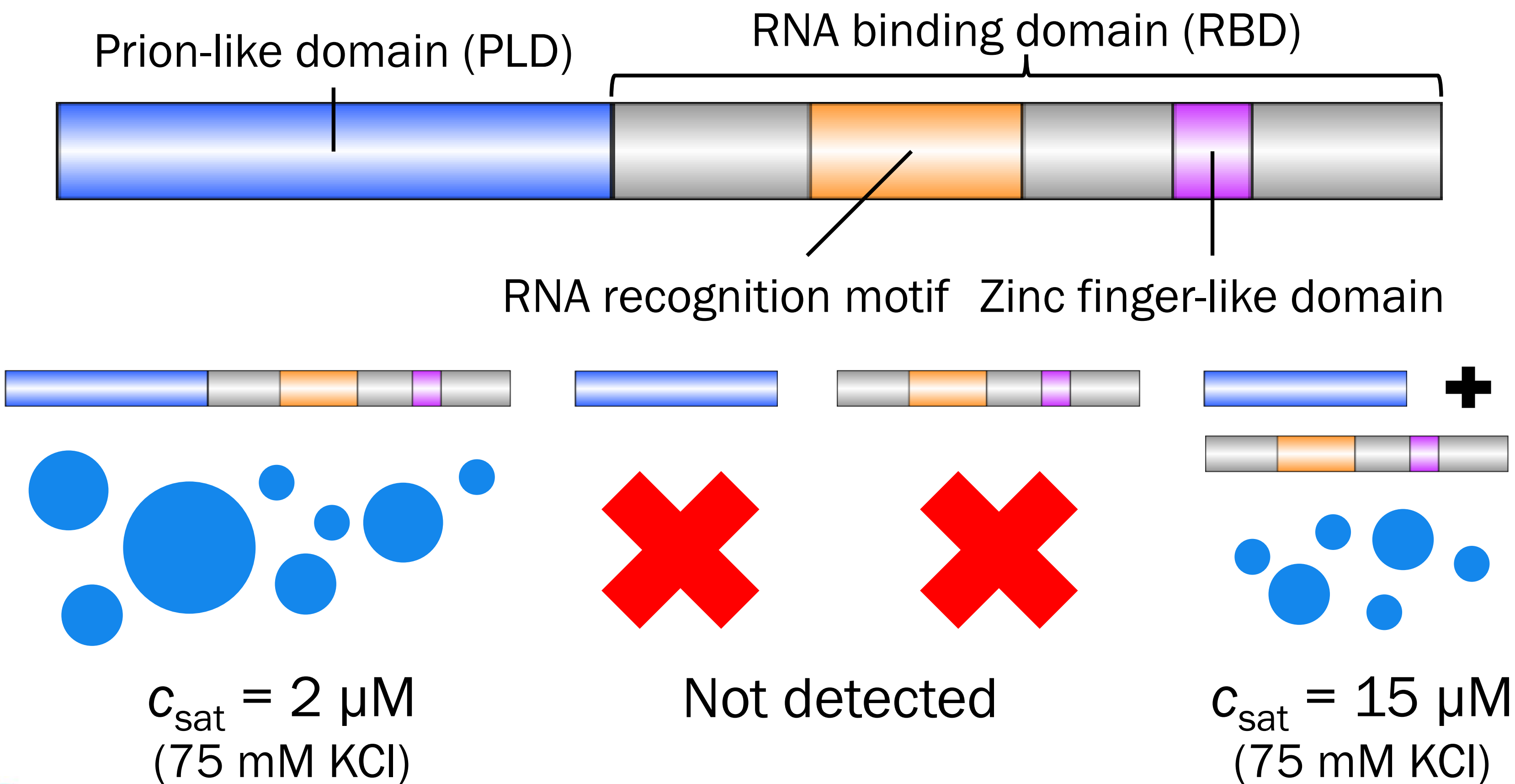


Interplay of Two Domains



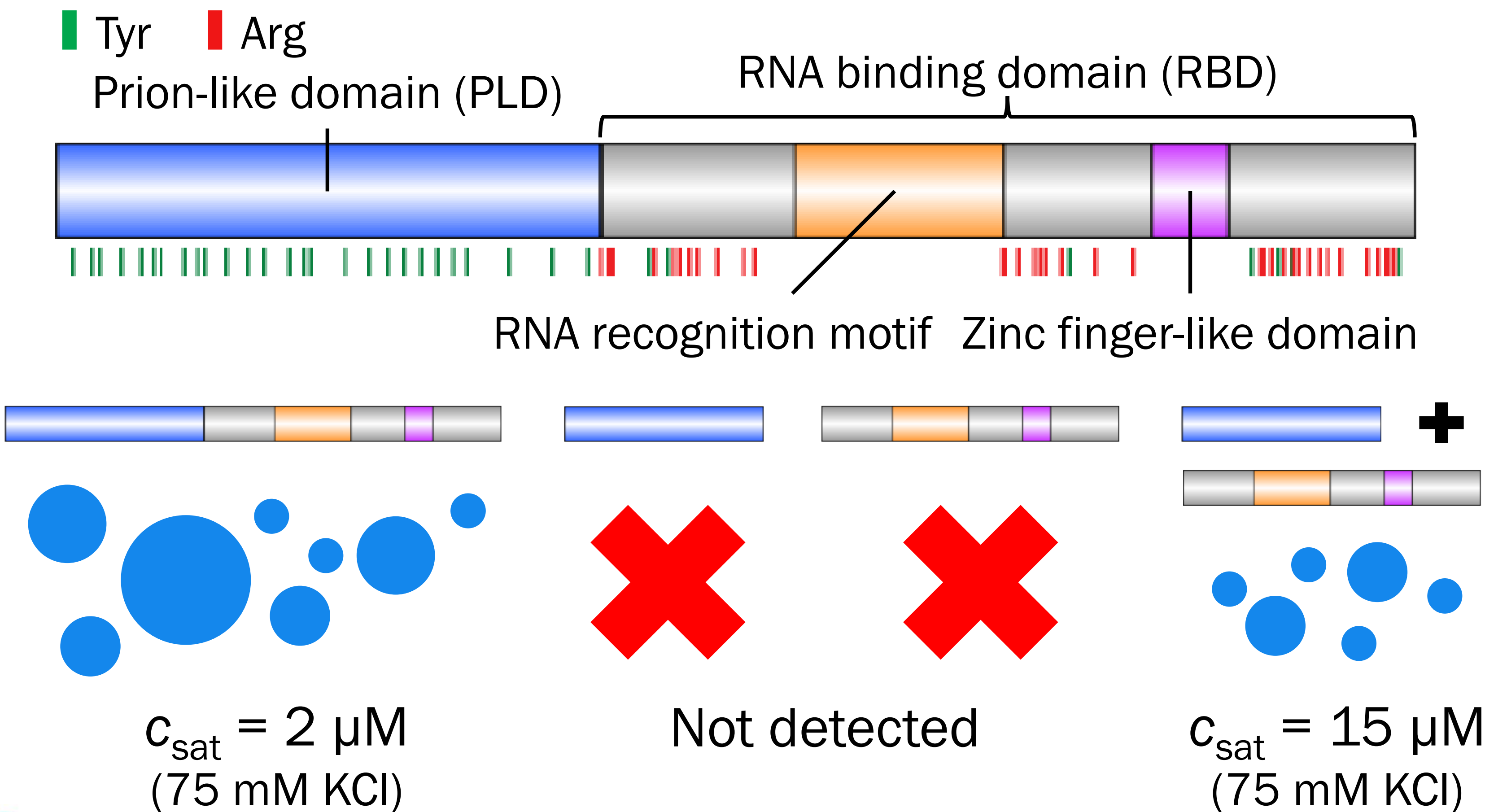
Wang *et al.*, ***Cell*** (2018)

Interplay of Two Domains



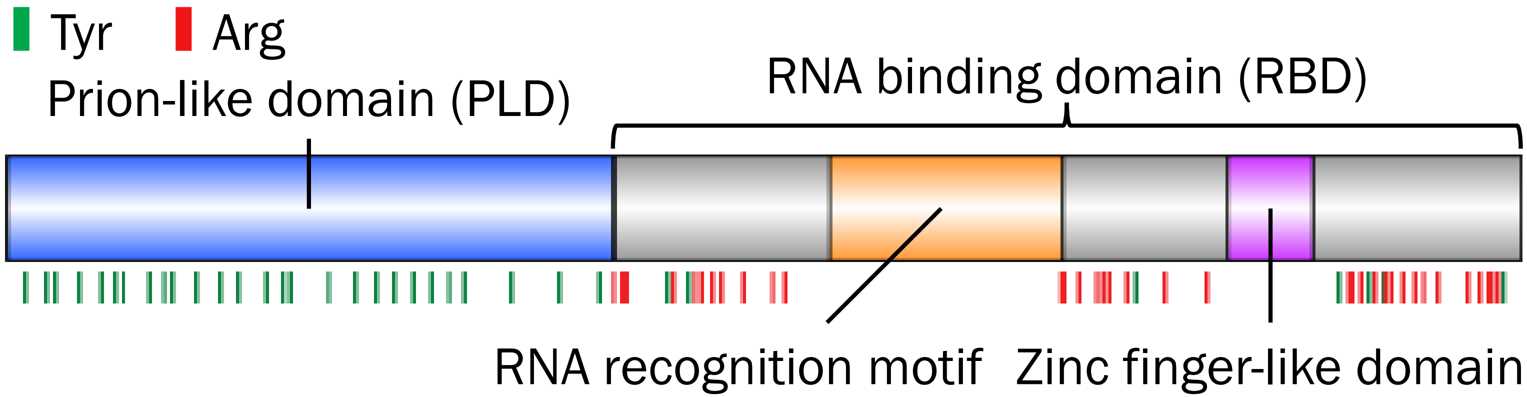
Wang *et al*., ***Cell*** (2018)

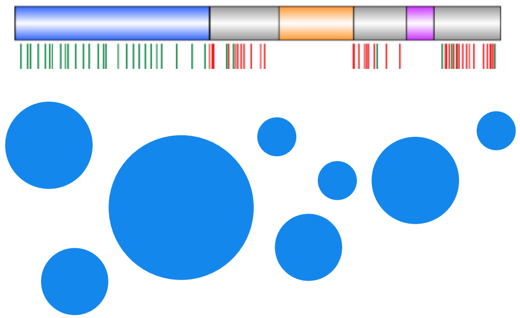
Tyrosines and Arginines?



Wang *et al.*, ***Cell*** (2018)

Tyrosines and Arginines?

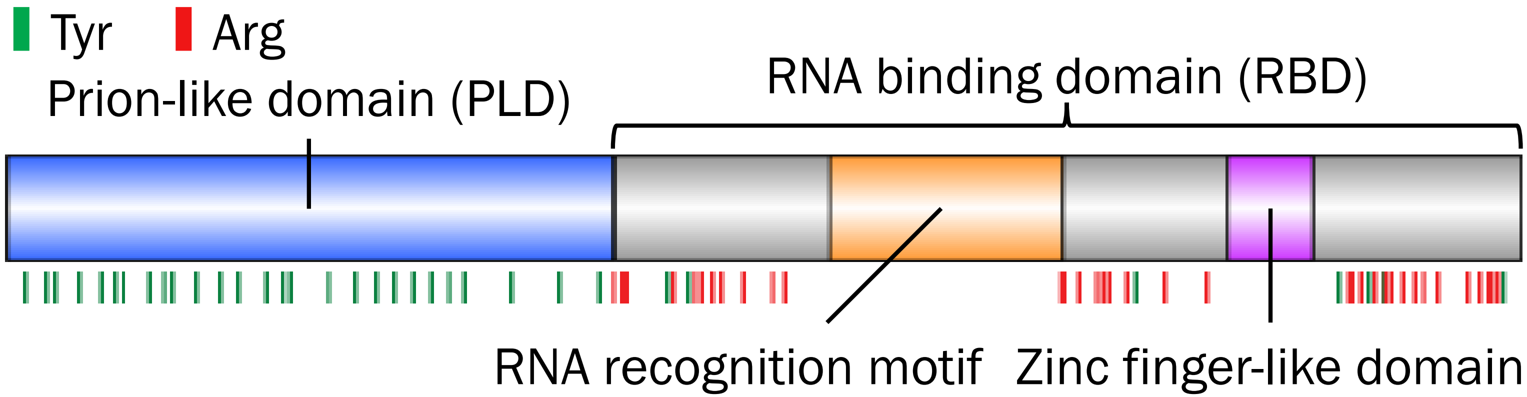


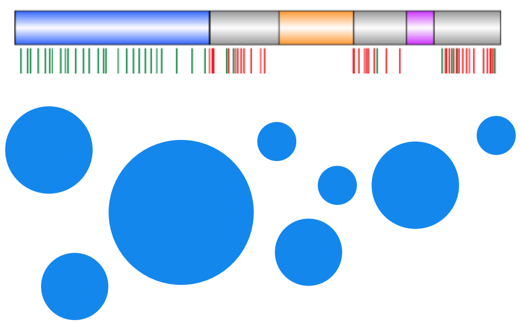


$c_{sat} = 2\ \mu M$
(75 mM KCl)

Wang *et al.*, ***Cell*** (2018)

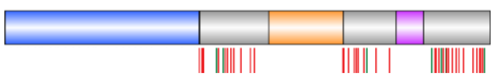
Tyrosines and Arginines?





c_{sat} = 2 μM
(75 mM KCl)

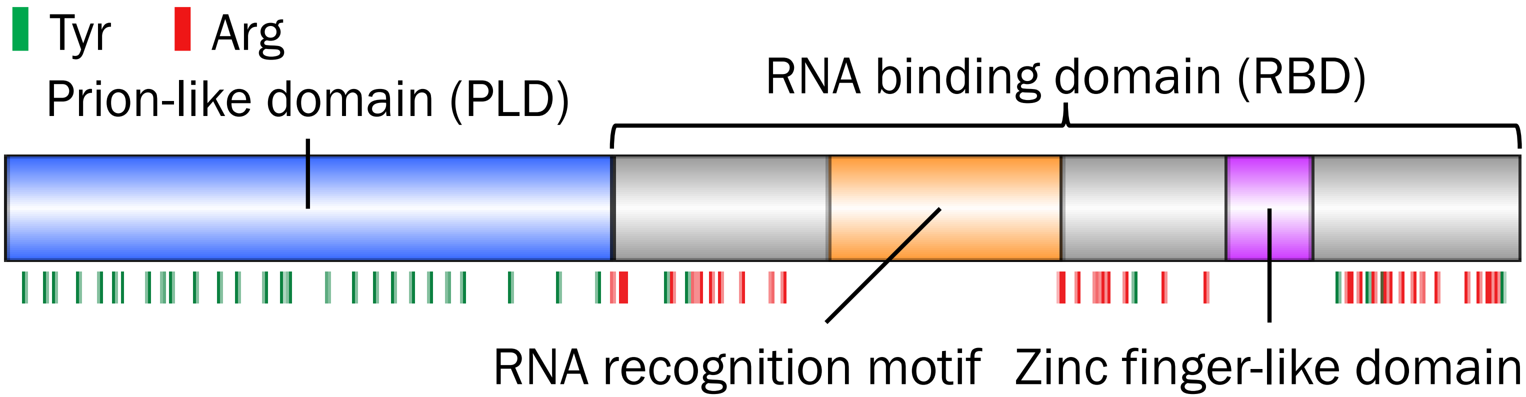
PLD Tyr → Ser

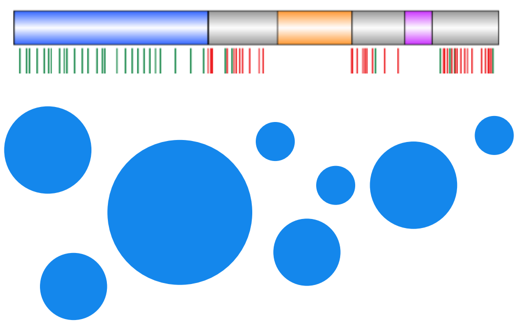


RBD Arg → Gly

Wang *et al.*, ***Cell*** (2018)

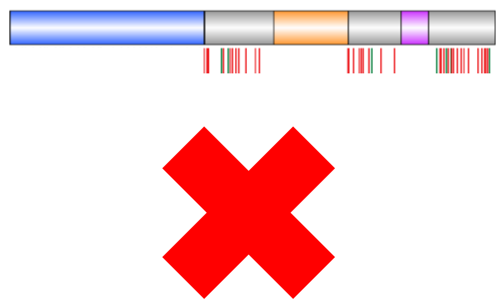
Tyrosines and Arginines?





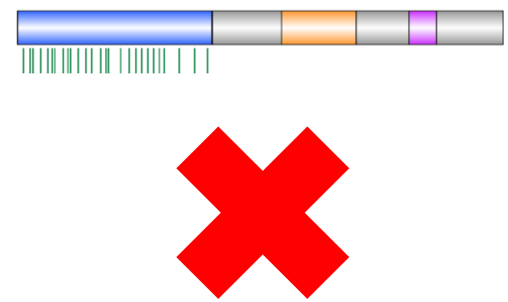
c_{sat} = 2 μM
(75 mM KCl)

PLD Tyr → Ser



Not detected

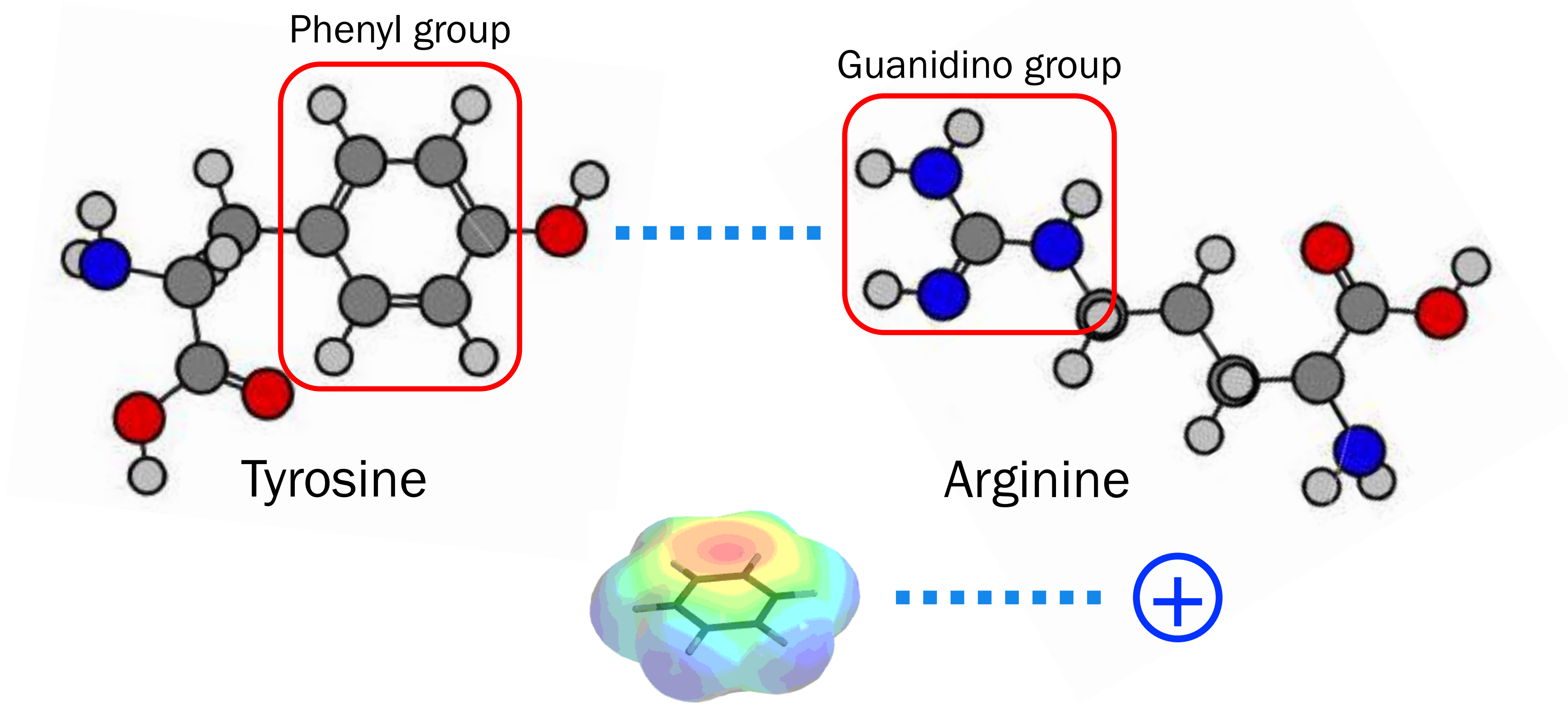
RBD Arg → Gly



Not detected

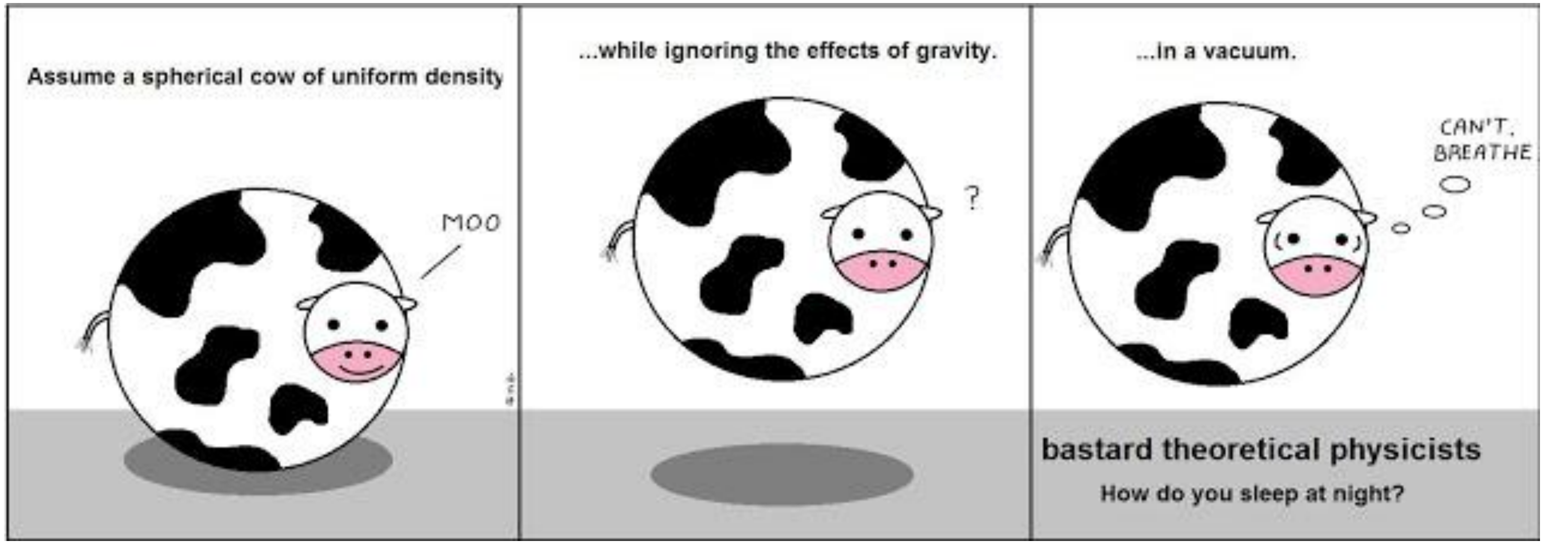
Wang *et al*., ***Cell*** (2018)

Chemistry of Tyr-Arg Interaction



Interaction between π system & positive charge (cation)

Spherical Cow in Vacuum

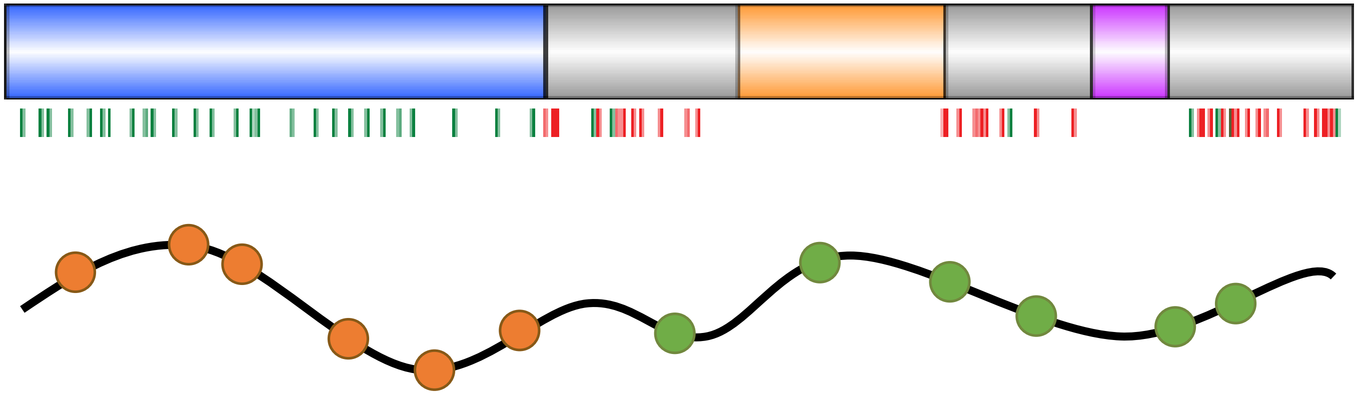


Let us start from the simplest model

Image: https://biotechdesign.artscinow.com/node/1156

“Stickers and Spacers” Framework

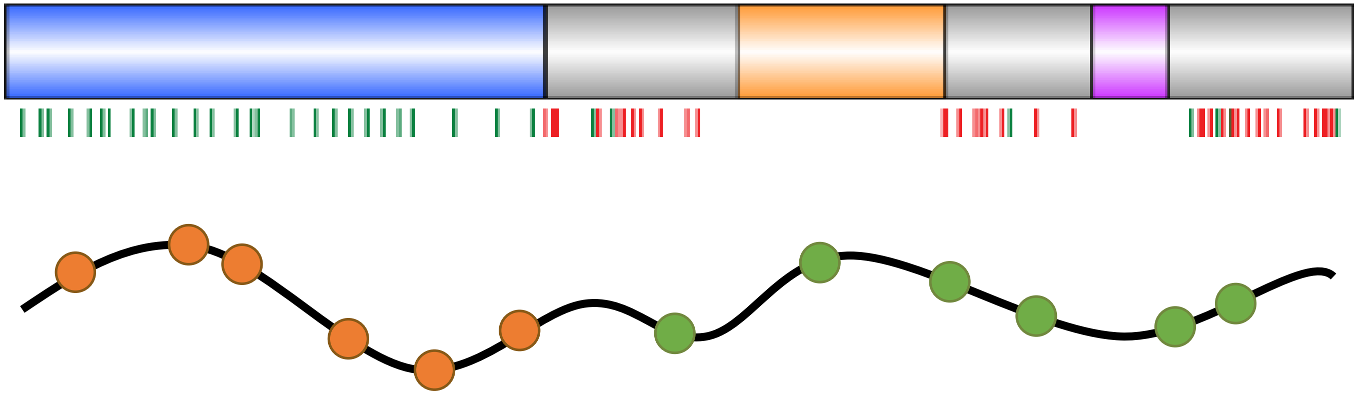
- Phase behaviors come from interplay between
  - Sticker residues: chain-chain interactions
  - Spacer residues: chain properties



Wang *et al.*, ***Cell*** (2018); Choi *et al.*, ***Annu Rev Biophys*** (2020)

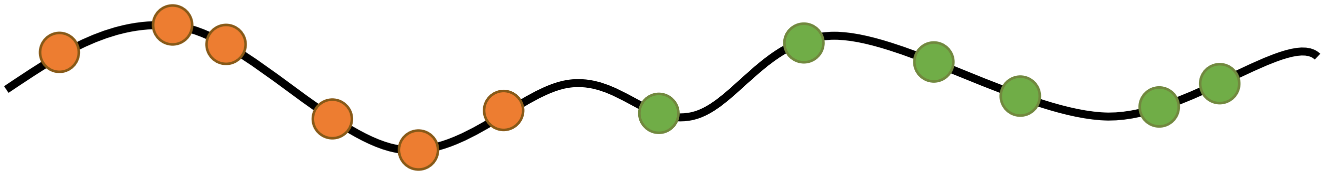
“Stickers and Spacers” Framework

- In the case of FUS
  - Sticker residues: Tyr and Arg
  - Spacer residues: other residues

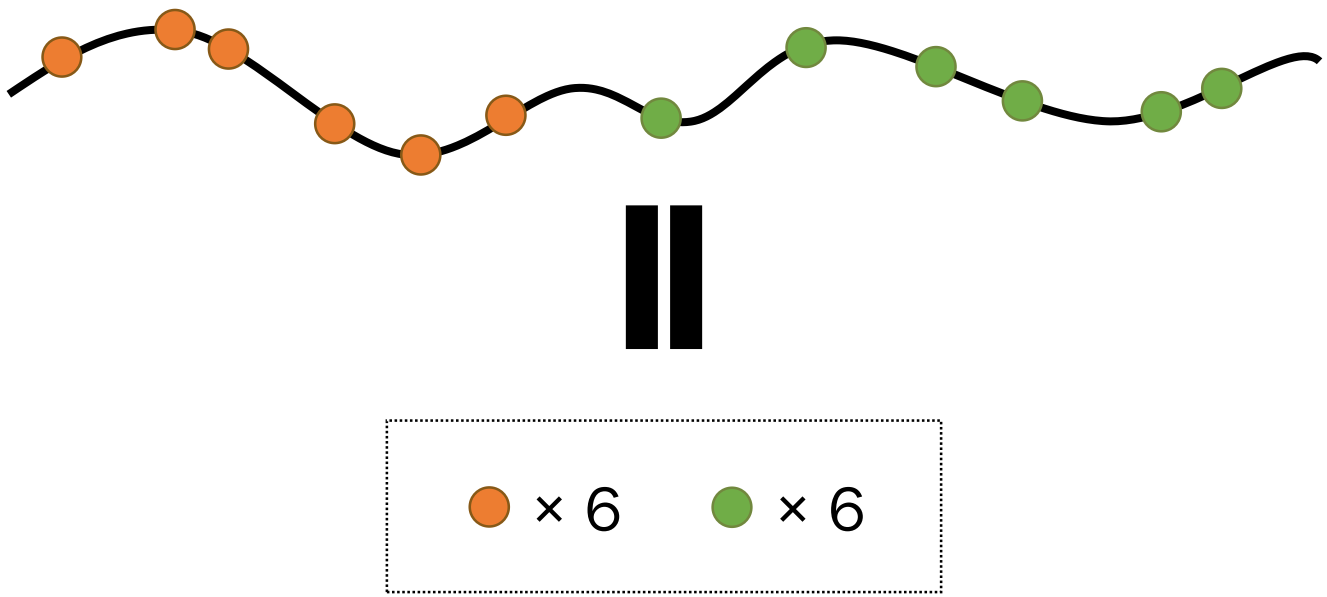


Wang *et al*., ***Cell*** (2018); Choi *et al*., ***Annu Rev Biophys*** (2020)

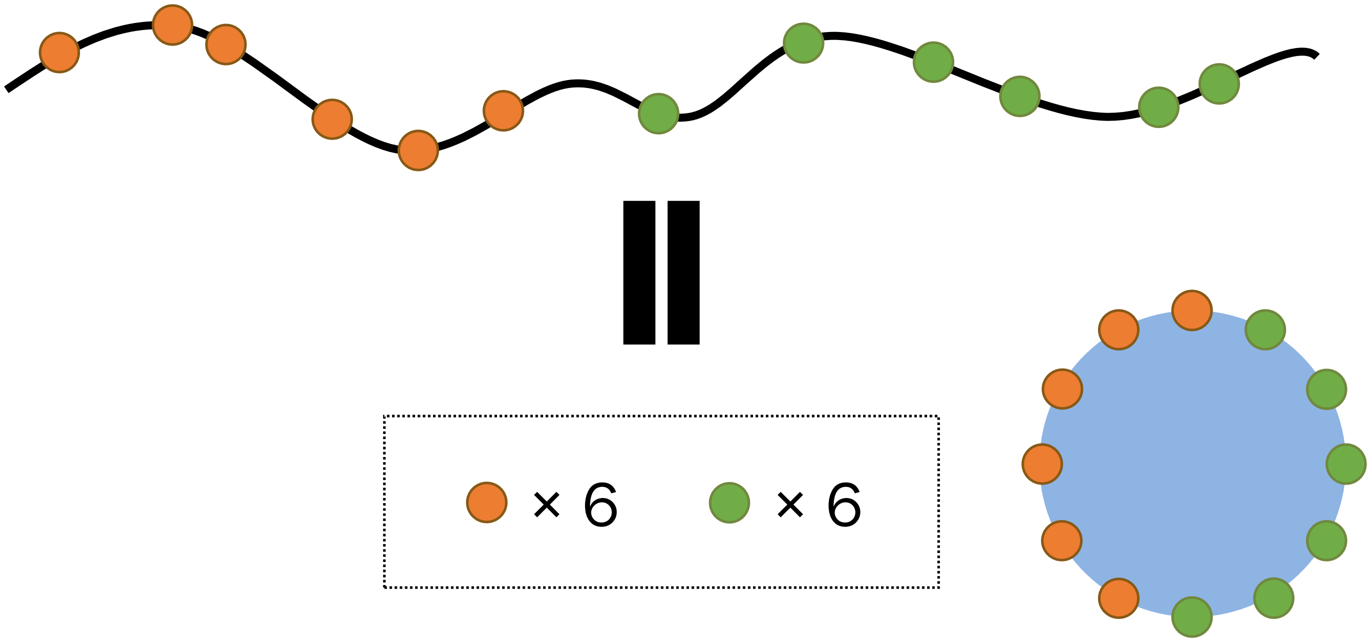
Suppose that the stickers are known …



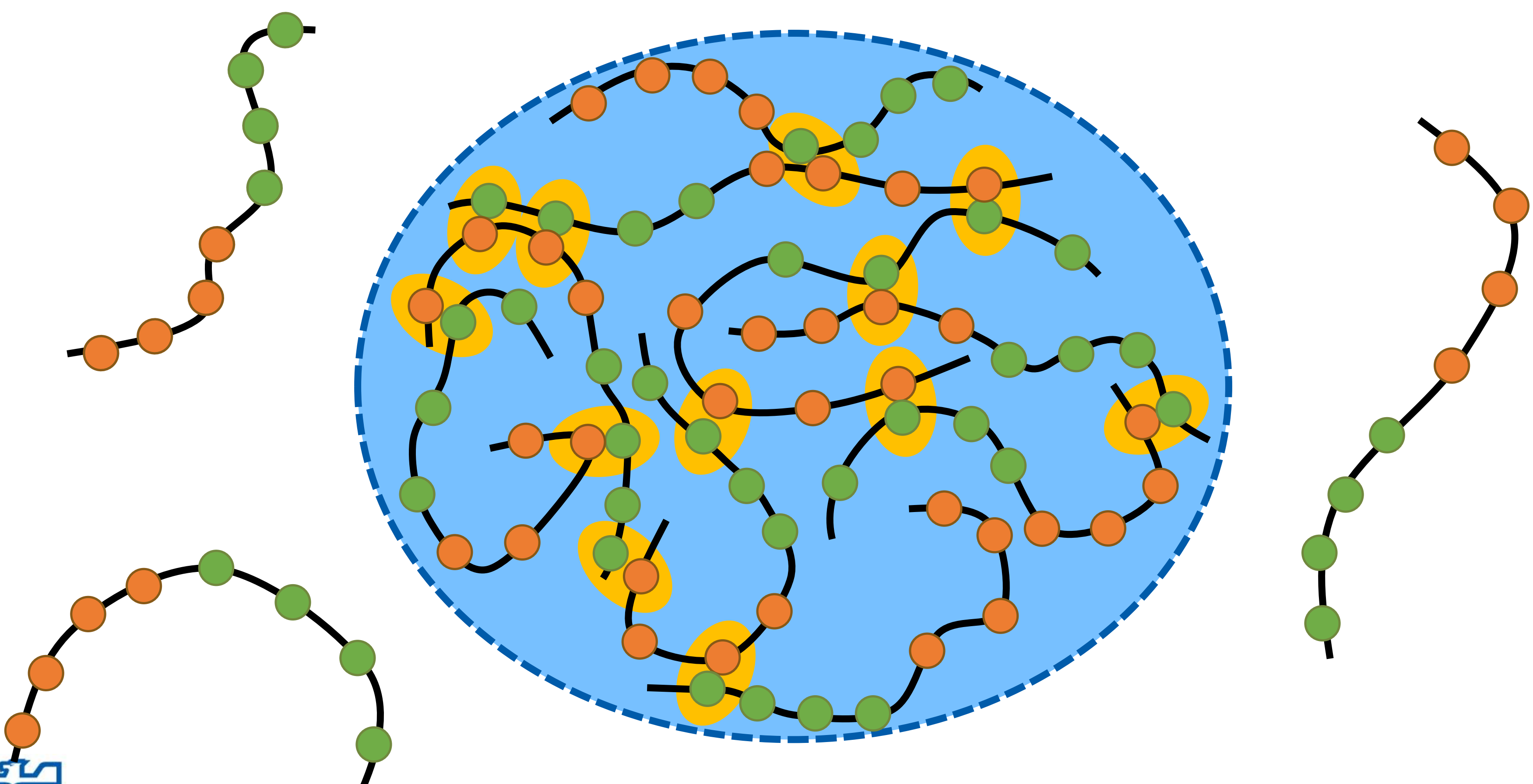
Mean-Field Approach



Mean-Field Approach



Network Formation for Condensation



Wang *et al.*, ***Cell*** (2018); Choi *et al.*, ***Annu Rev Biophys*** (2020)

Mean-Field Prediction

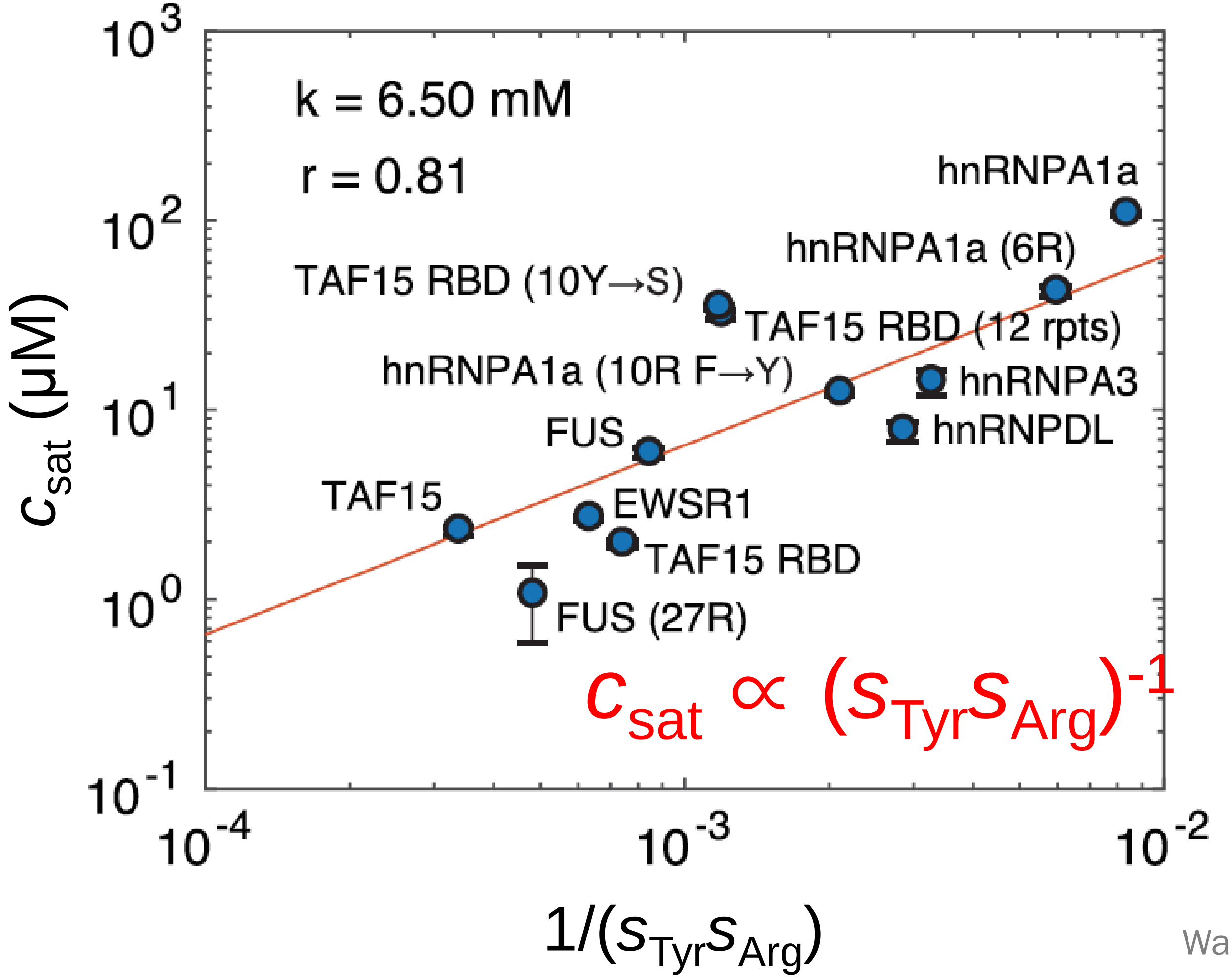
- Two types of stickers (A & B)
- Only A:B interactions allowed

s_A: number of A-stickers & s_B: number of B-stickers

$$c_{sat} \propto (s_A s_B)^{-1}$$

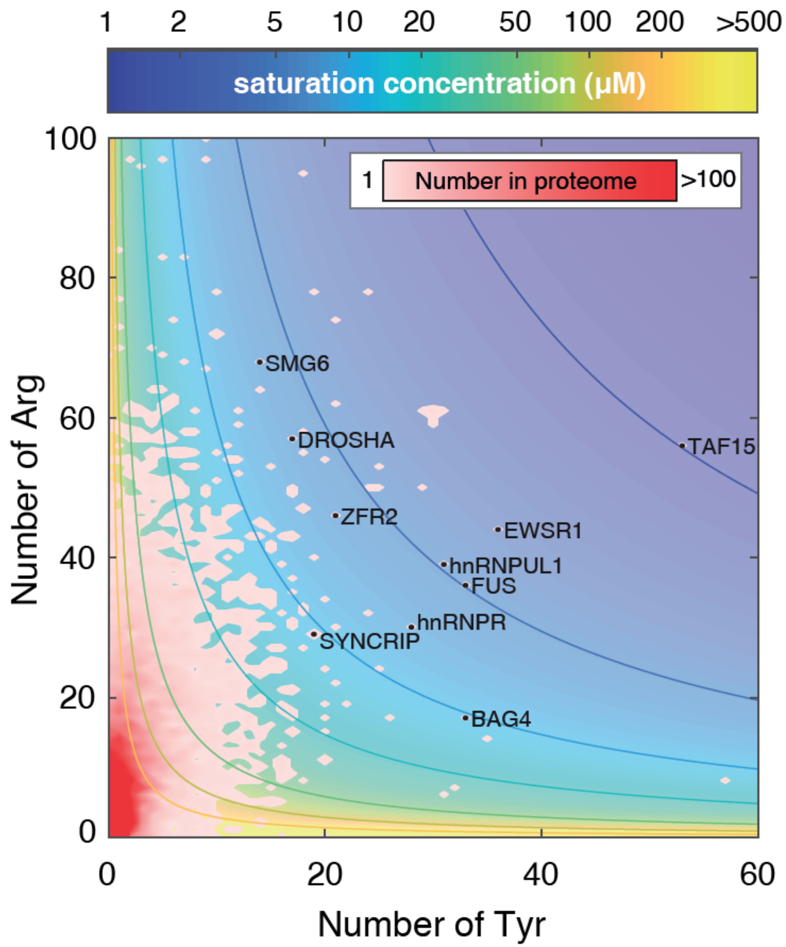
Wang *et al*., ***Cell*** (2018); Choi *et al*., ***Annu Rev Biophys*** (2020)

Theory Explains Experiment



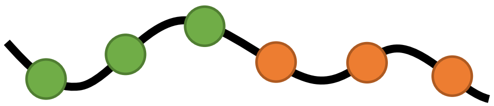
Wang *et al.*, ***Cell*** (2018)

Theory also Gives Predictions

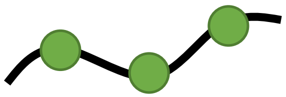


Wang *et al.*, ***Cell*** (2018)

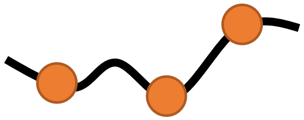
Separation Effect



vs.



+

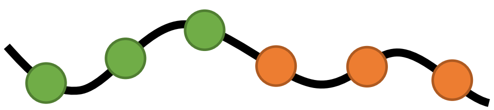


c_{sat} = 2 μM
(FUS, 75 mM KCl)

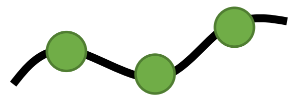
c_{sat} = 15 μM
(FUS, 75 mM KCl)

Choi *et al*., unpublished

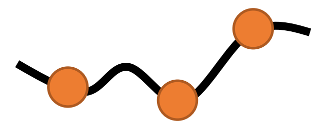
Qualitative Explanation



vs.



+



c_{sat} = 2 μM
(FUS, 75 mM KCl)

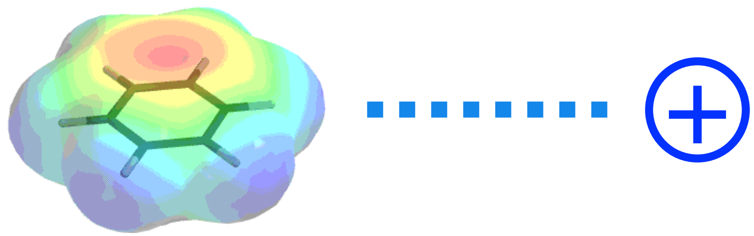
c_{sat} = 15 μM
(FUS, 75 mM KCl)

$$c_{\text{sat}} \approx \frac{1}{2\lambda s_{\text{A}} s_{\text{B}}}$$

$$c_{\text{sat}} \approx \frac{1}{\lambda s_{\text{A}} s_{\text{B}}}$$

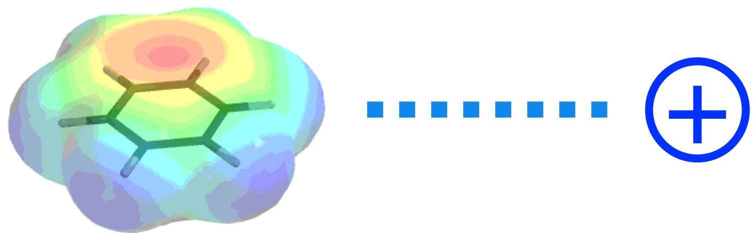
Choi *et al.*, unpublished

Chemistry of Stickers

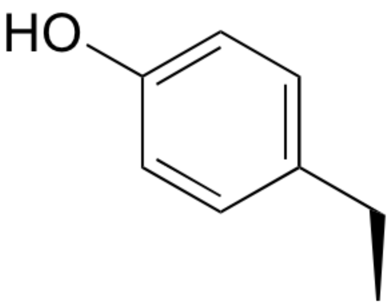


Interaction between π system & positive charge (cation)

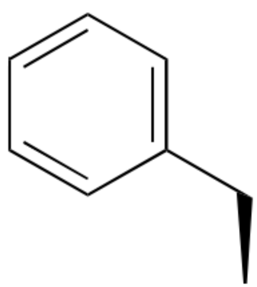
Chemistry of Stickers

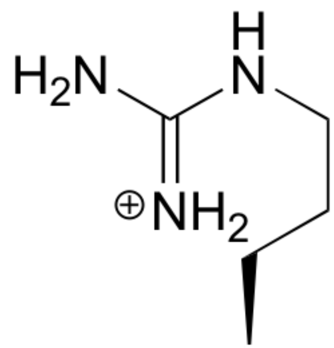


Interaction between π system & positive charge (cation)

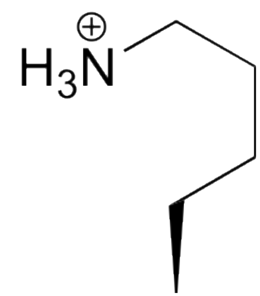


vs.





vs.



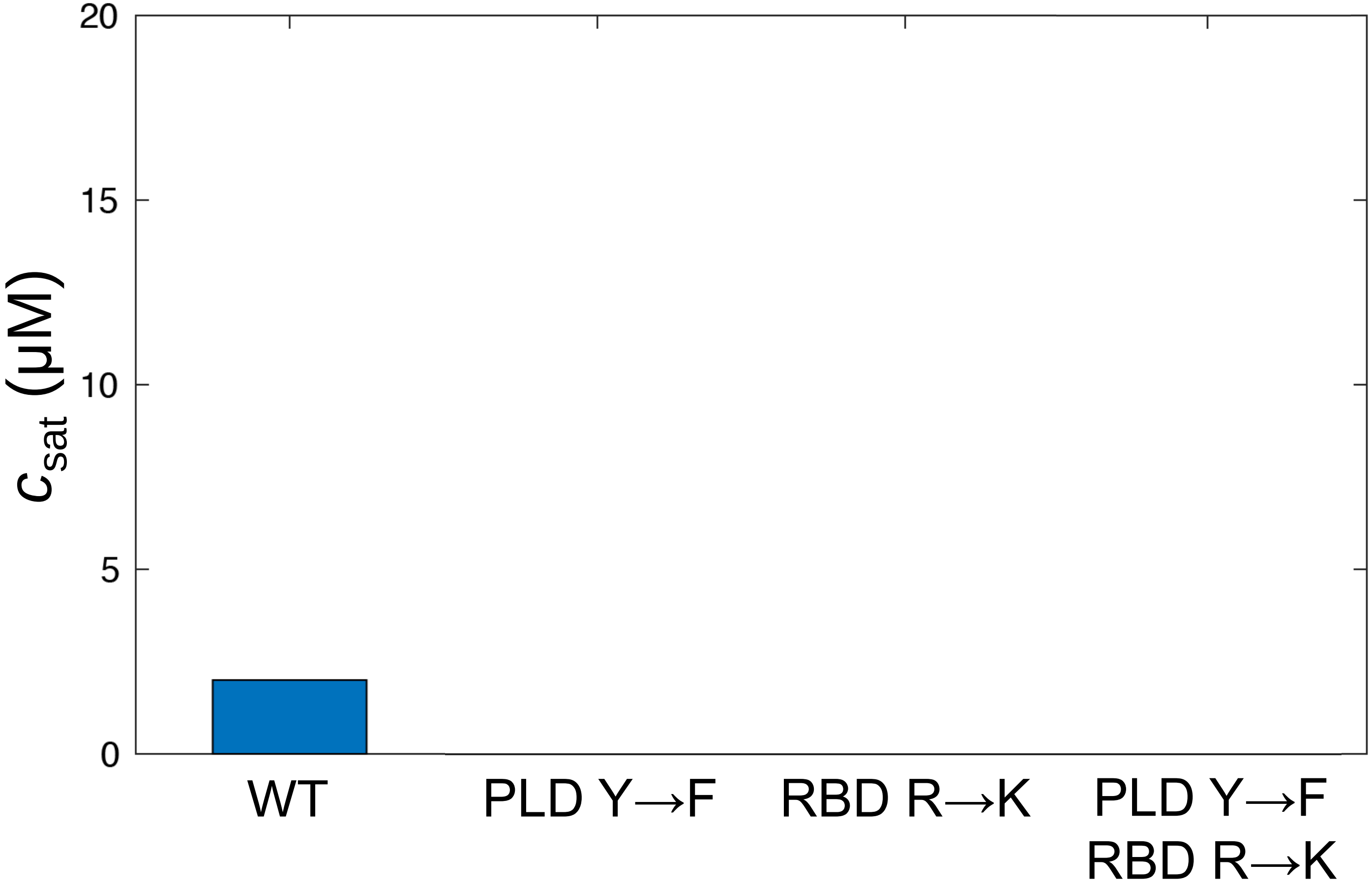
Tyr (Y)

Phe (F)

Arg (R)

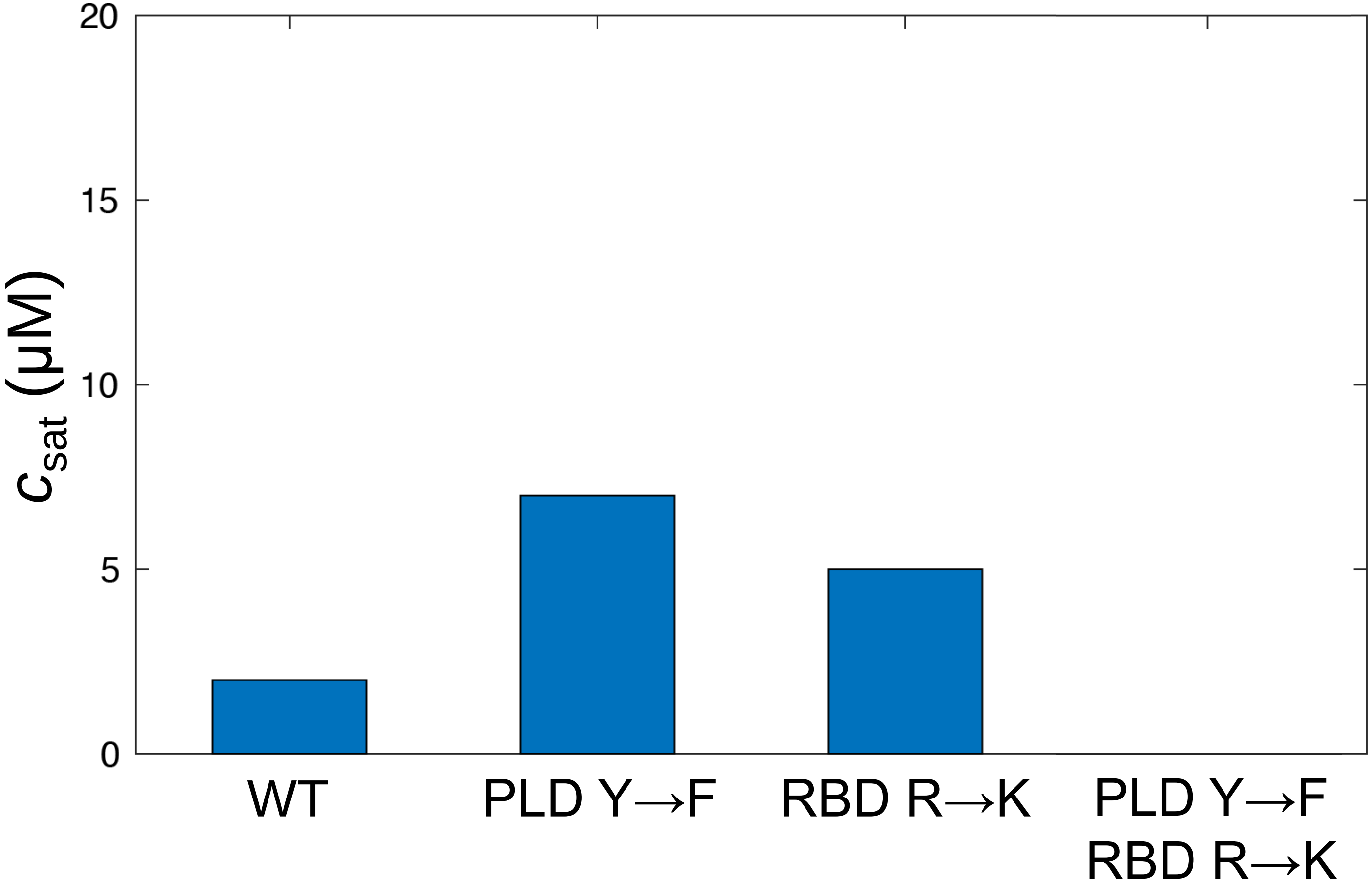
Lys (K)

FUS Mutants (75 mM KCl)



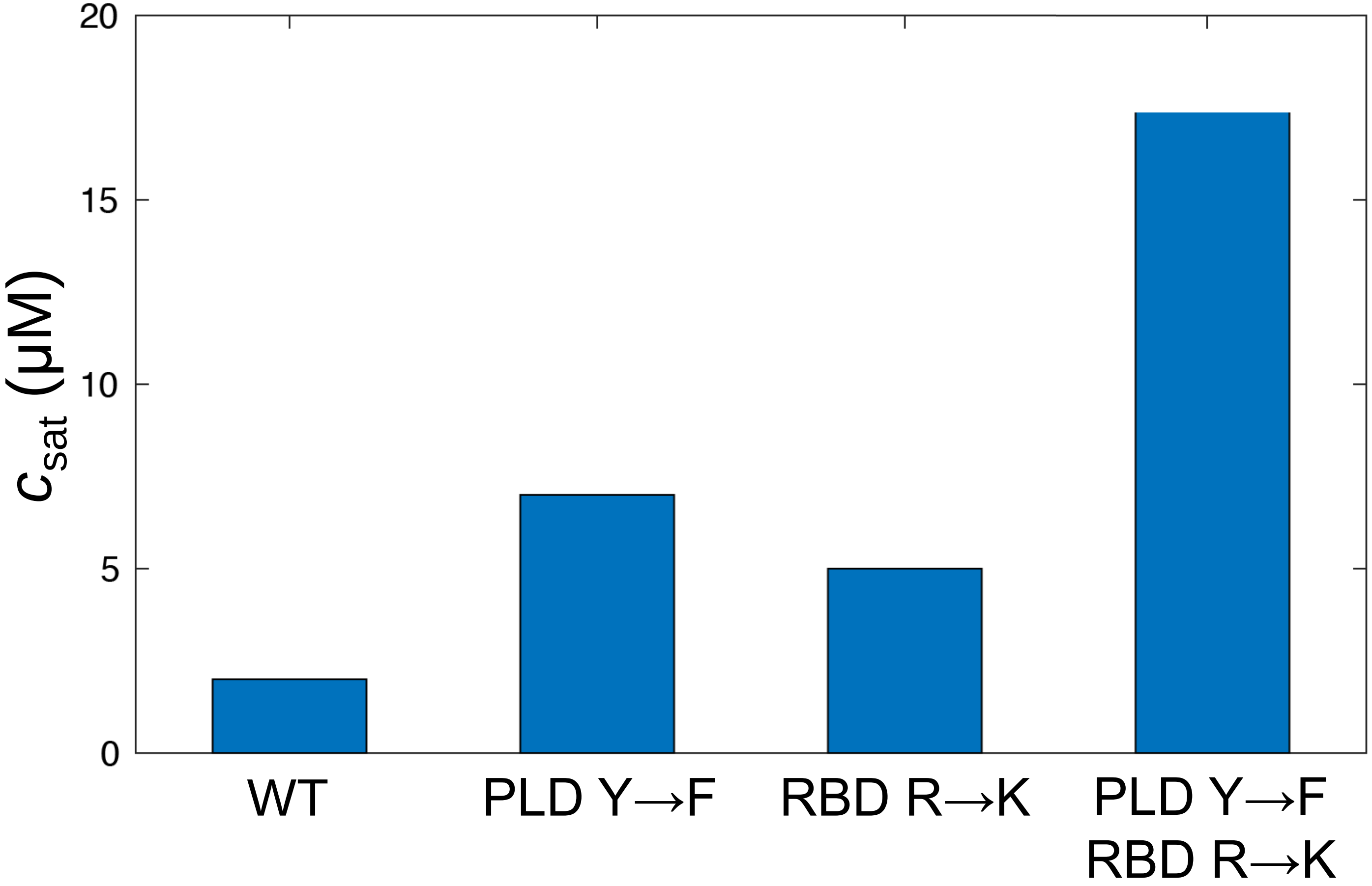
Wang *et al.*, ***Cell*** (2018)

FUS Mutants (75 mM KCl)



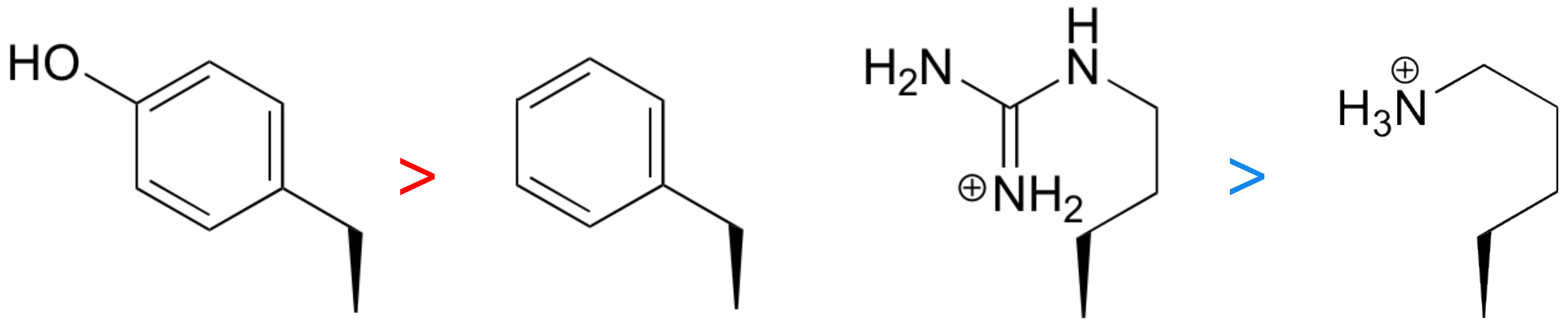
Wang *et al.*, ***Cell*** (2018)

FUS Mutants (75 mM KCl)



Wang *et al*., ***Cell*** (2018)

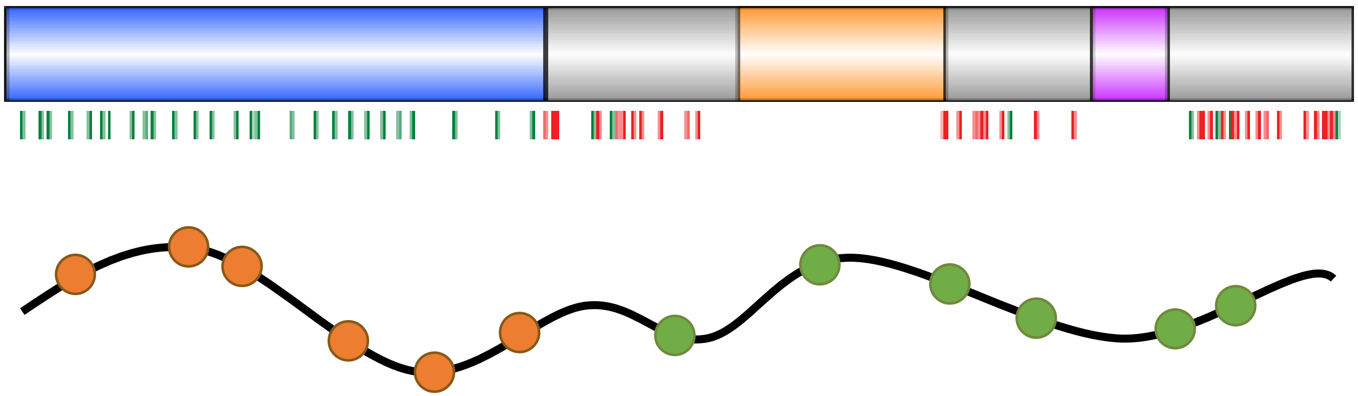
Sticker Strengths



Tyr (Y) Phe (F) Arg (R) Lys (K)

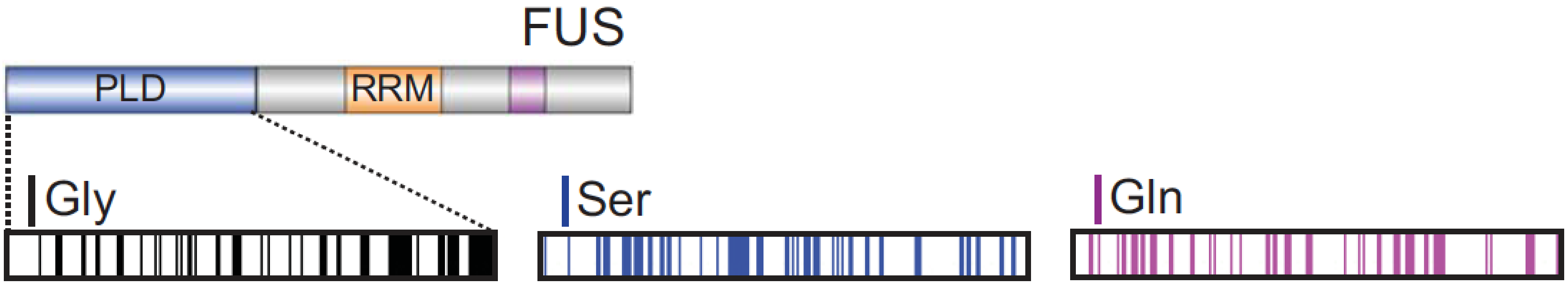
“Stickers and Spacers” Framework

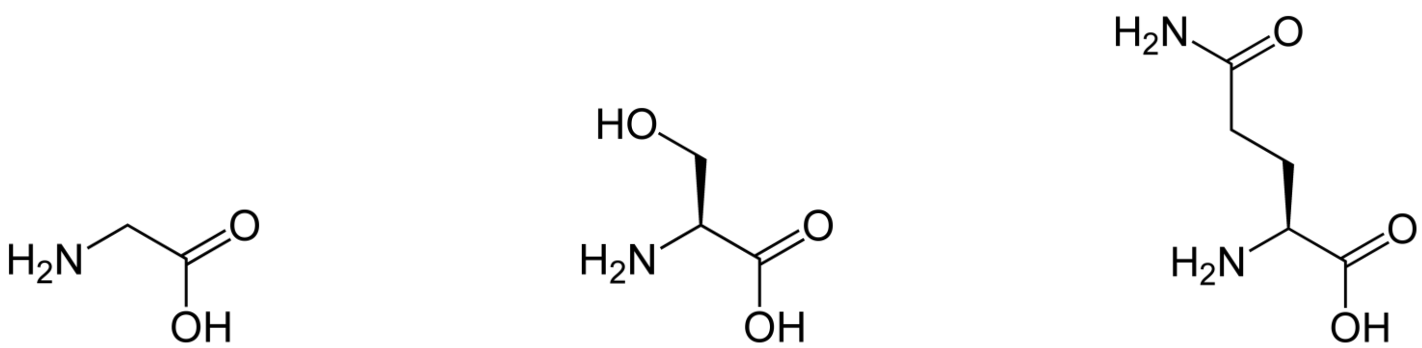
- Phase behaviors come from interplay between
  - Sticker residues: chain-chain interactions
  - Spacer residues: chain properties



Wang *et al*., ***Cell*** (2018); Choi *et al*., ***Annu Rev Biophys*** (2020)

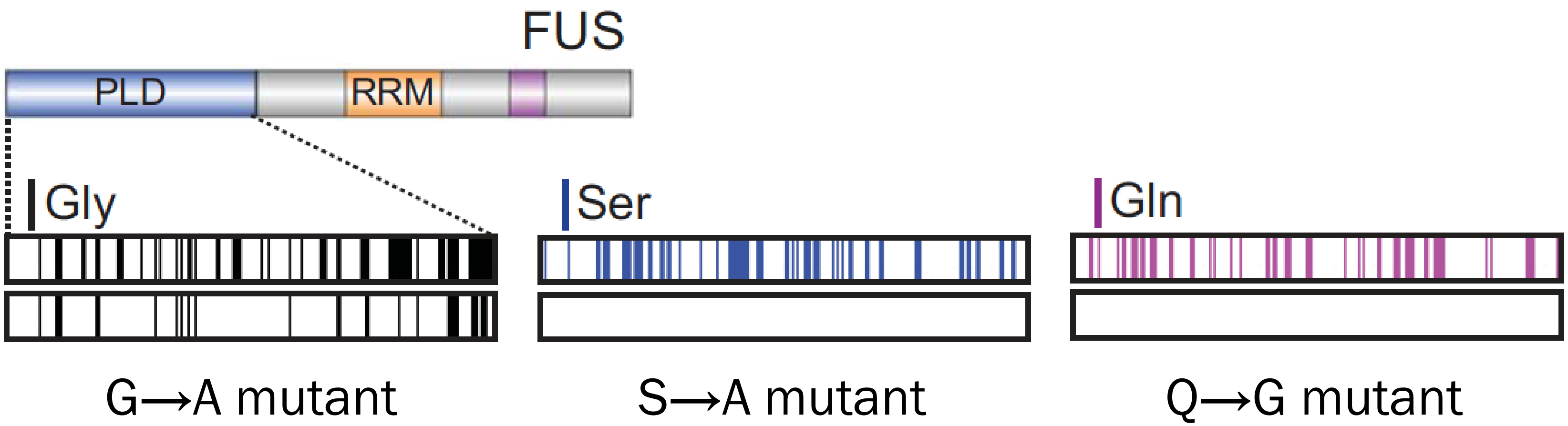
“Spacer” Candidates





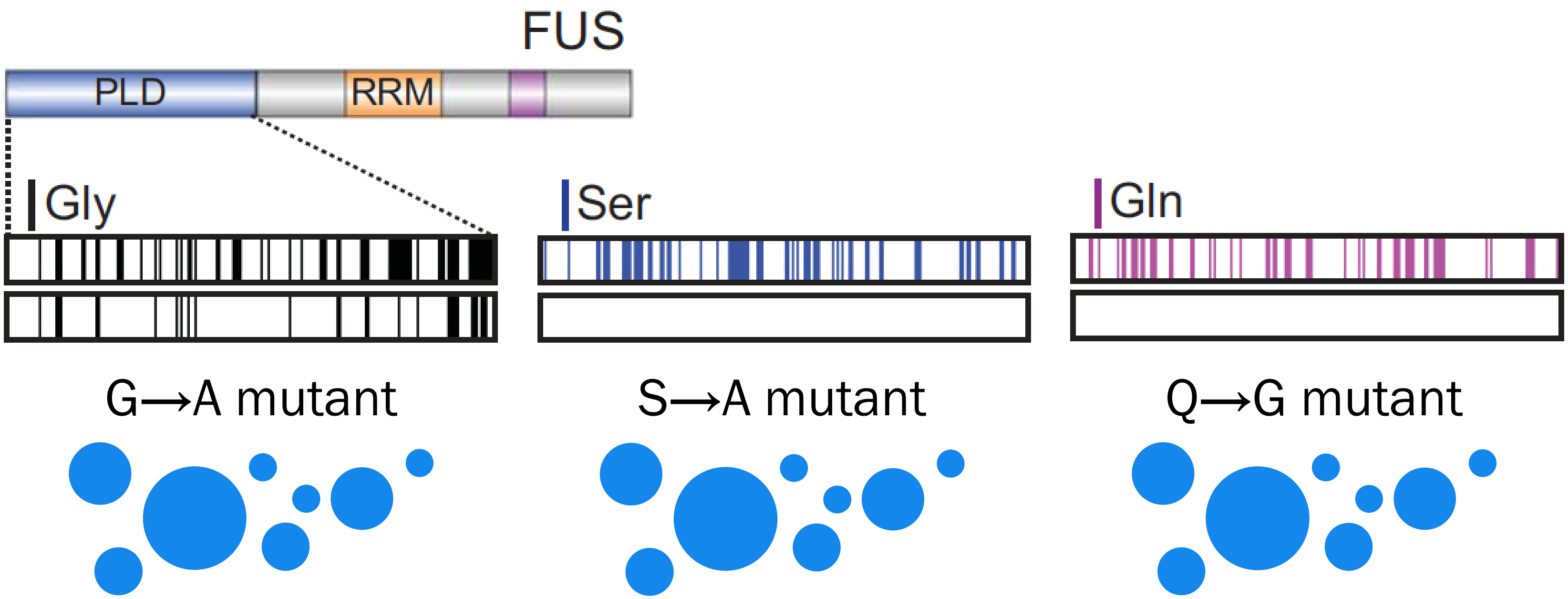
Wang *et al.*, ***Cell*** (2018)

Preparation of Mutants



Wang *et al.*, ***Cell*** (2018)

They All do not Change c_{sat}



c_{sat} = 1-2 μM (at 75 mM KCl)

Wang *et al*., ***Cell*** (2018)

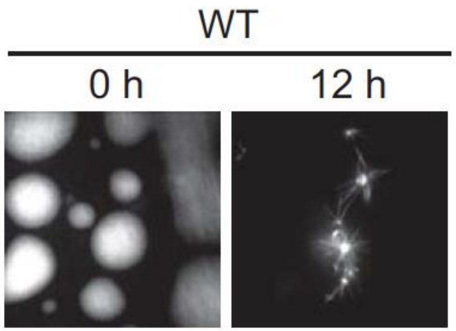
What do Spacers do? Case of G

Wildtype @ 120 min

G⟶A mutant @ 40 min

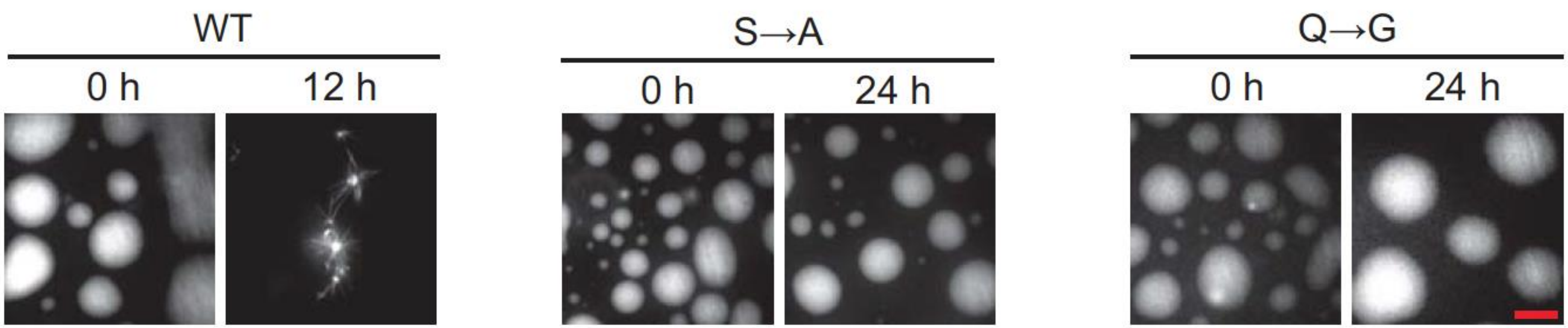
Wang *et al*., ***Cell*** (2018)

What do Spacers do? Case of S and Q



Wang *et al.*, ***Cell*** (2018)

What do Spacers do? Case of S and Q



Bar: 5 μm

Wang *et al*., ***Cell*** (2018)

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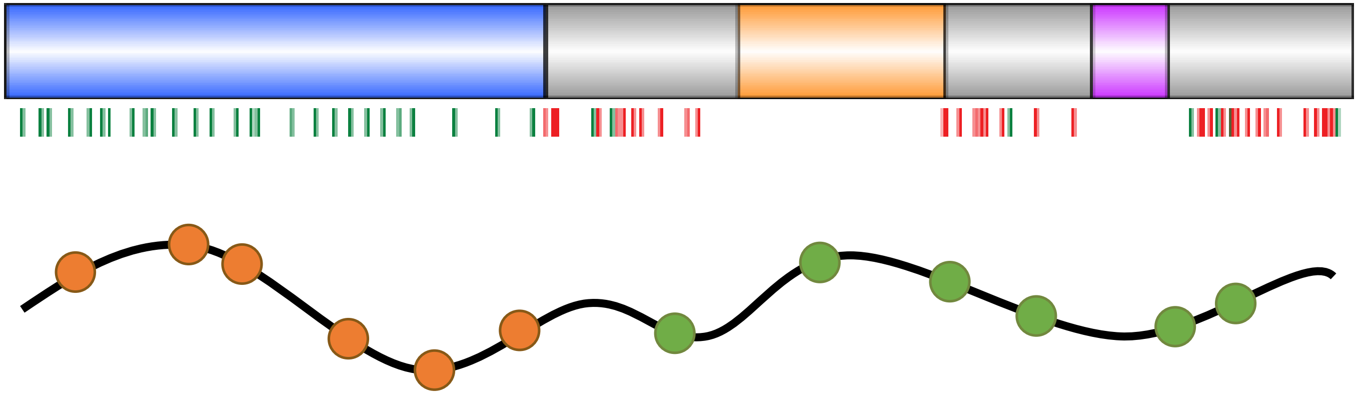

Wang *et al*., ***Cell*** (2018); Choi *et al*., ***Annu Rev Biophys*** (2020)